



Dynamics of batch protein precipitation

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ABSTRACT

Precipitation is a valuable alternative to recovery processes like chromatography. While the process is a plausible alternative, without experimentation, there is a lack of sufficient understanding of how process operating conditions lead to expected chord length distributions (also referred to as particle size distributions). In this work, a modeling framework is proposed through which, first, the system is specified, then the population balance model (PBM) is constructed. Data is then gathered from experiments and subsequently processed. Afterwards, an optimization problem is defined to regress the PBM parameters to the experimental data. Finally, a surrogate model correlating the PBM parameters with process operating conditions is developed. Thereby, avoiding future experiments and regression of the parameters. The application of the modeling framework is demonstrated using 15 batch experiments for lysozyme precipitation and particle size distribution measurements using focused beam reflectance measurement (FBRM). The current approach helps identify the correlations between process operating conditions and process model (PBM) parameters such as maximum collision efficiency and breakage rate coefficient, which in turn helps in the estimation of the population distribution in the system. The work also proposes the limitations with modeling and data acquisition that pave the way for future research.

1. Introduction

Recovery of a biopharmaceutical molecule, such as a protein, usually includes a capture chromatography step, and two chromatographic polishing steps, followed by ultrafiltration/diafiltration. Downstream processing is especially significant as manufacturing costs typically account for about 20% of total product sales, with downstream processing alone representing about 75% of total manufacturing costs (Decker et al., 2023). To overcome the limitations associated with chromatography (see Fig. 1), such as high costs of resins, diffusion issues, and capacity constraints, there is a growing interest in exploring alternative approaches known as “Anything but Chromatography” methodologies (Roque et al., 2020). Currently, nonchromatographic techniques for downstream processing are experiencing renewed interest, including precipitation methods (dos Santos et al., 2017).

Precipitation can be utilized as a protein recovery method that is economical, sustainable, and amenable to continuous processing (dos Santos et al., 2017; Walla et al., 2023), and can produce high yields of a pure, functional protein. An early demonstration of this principle is the Cohn precipitation process which is still in use for the separation of blood products (Cohn et al., 1946). The process utilizes ethanol

and salts, and manipulates pH and temperature to sequentially remove proteins of therapeutic value from whole plasma (Cohn et al., 1946).

There are two separation strategies available for recovering biologically produced molecules such as proteins, by precipitation. The first strategy involves precipitating side components, such as deoxyribonucleic acid (DNA) and host cell proteins (HCP), while keeping the target protein in solution to avoid subsequent protein dissolution and degradation issues. The second strategy involves using precipitation to isolate the target protein itself, which is more convenient because optimal parameters can be selected for one specific target component. Both strategies involve adding a precipitating agent to the protein-containing solution. The first strategy causes the impurities such as proteases (enzymes that break down proteins) to aggregate and become insoluble, whereas the second strategy causes the protein of interest to precipitate, while other components remain in the soluble phase. The resulting protein aggregates can be easily separated from the solution by centrifugation or filtration. The first strategy is a purification method while the second strategy is a recovery method. In this work we focus on recovery. This strategy can significantly enhance the recovery of the enriched precipitated protein (Burgess, 2009; Gilleskie et al., 2021).

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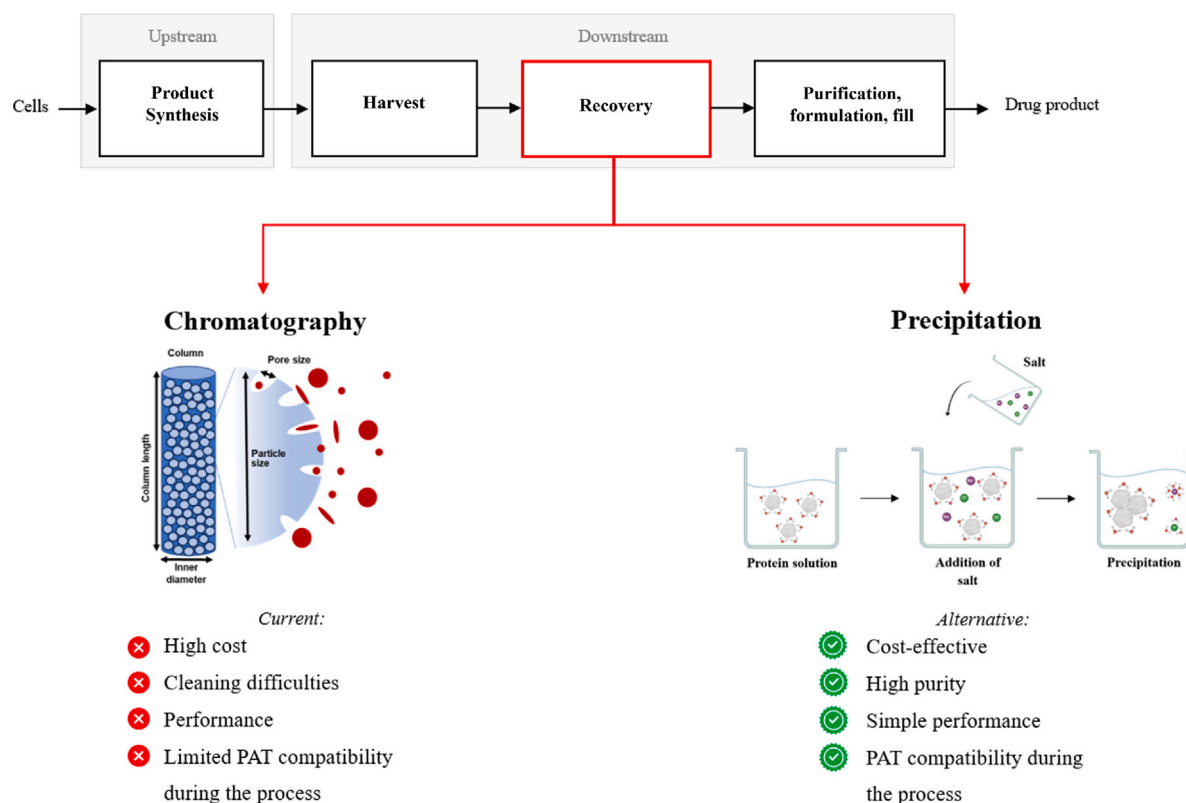


Fig. 1. Overview of the conventional versus alternative purification methods in biopharmaceutical manufacturing.

Precipitation does not require a complex equipment setup compared to chromatography, a continuous stirred tank (CST) or a tubular vessel with a filter for solid liquid separation works well. It can deal with high process volumes and rising titers and can have shorter process times than chromatography. Furthermore, for applications on a large scale, the volume of the CST is easily scalable, and continuous operation is feasible (Hammerschmidt et al., 2016; Burgstaller et al., 2019). Selective precipitation steps can help skipping two of the usual three chromatography steps, and it is shown that the precipitation offers cost reductions at all stages of the life cycle of a therapeutic antibody (Hammerschmidt et al., 2014).

Still challenges to implementation of alternative separation methods in biopharmaceutical manufacturing remain apparent. However, recent advances in upstream process intensification and adoption of new technologies enabling continuous manufacturing have successfully increased the titers of even complex molecules like monoclonal antibodies making these alternative methods potentially more attractive (Müller et al., 2022). There is also a lack of general developmental knowledge for precipitation processes, and concerns over regulatory approval due to limited process data generated by the precipitation process. In this work, we demonstrate a use-case towards recovery of a selected protein by precipitation, using key factors that affect its precipitation from the protein solution.

Several factors affect protein precipitation, including pH of the solution, salt concentration, temperature, agitation, protein concentration, presence of co-solvents, and buffer solutions. Variation in pH impacts protein solubility and precipitation by altering the charge distribution on the protein surface. Salt concentration and salt type, as outlined in the Hofmeister series, significantly influences protein stability and solubility due to ions kosmotropic or chaotropic properties. Temperature affects the balance between hydrophobic interactions and other forces that maintain protein solubility. Agitation improves precipitation efficiency by promoting the mixing of precipitants into protein solutions and increasing the probability of collision (Sommer

et al., 2015). Protein concentration determines the solubility limit and the likelihood of precipitation (Burgess, 2009). The presence of co-solvents and buffering agents also affects protein solubility and impacts interactions crucial for protein precipitation (dos Santos et al., 2017). These factors such as salt and protein concentration, agitation, etc., collectively shape the outcome of protein precipitation processes by modifying solubility, stability, and protein-solution interactions and a unique set of values of these factors will affect proteins in a different way.

1.1. Batch processing and monitoring solutions

Despite the inherent simplicity of batch processing, no established standard for protein precipitation conditions currently exists, primarily because the unique and complex structures of individual proteins lead to distinct biological characteristics. These processes are highly sensitive to various inputs, such as shear forces, salt concentration, and protein concentration. Altering any of these parameters can significantly impact the success of protein recovery by precipitation. Therefore, precise control of process operating conditions is crucial for the successful execution of these processes.

Focused Beam Reflectance Measurements (FBRM) and in situ microscopy via Process Video and Measurement (PVM) microscopes are available tools for monitoring precipitation and crystallization operations (Zhang et al., 2024; Walla et al., 2023). In situ monitoring tools are increasingly preferred because of their ability to provide near real-time information without the delays and additional labor associated with ex situ particle analysis techniques. Additionally, in situ methods help mitigate potential artifacts caused by sample preparation, such as changes in precipitate conformation due to dilution or alterations in ionic strength that is often necessary for at-line or off-line sampling. Modern advances allow for the analysis of larger particle populations using near real-time particle imaging techniques. These methods typically involve capturing images with a confocal microscope

and highspeed camera, followed by processing with segmentation algorithms to identify and extract various particle properties for each particle (Patchigolla and Wilkinson, 2009). Although techniques such as laser diffraction or electrical sensing zones may not be suitable to accurately determine particle size and shape distribution, microscopy coupled with image analysis, whether *ex situ* or *in situ*, is widely used (De Anda et al., 2005).

FBRM provides the chord length distribution (CLD) of a particle population, operates by analyzing back-scattering from a rotating monochromatic laser beam. Its ability to function even in concentrated slurries eliminates the need for additional dilution steps or external sampling, making it suitable for online monitoring. However, the effectiveness of FBRM is highly dependent on the optical properties of the surfaces of the particles. It provides highly sensitive real-time tracking of particle size, number, and shape changes without the need for off-line sampling. It also detects key process events such as nucleation, growth, and dissolution, for fingerprinting particle system distributions. However, its ability to directly track crystal size is limited to spherical crystals, as the measured CLD corresponds directly to the crystal size distribution (CSD) only in these cases. For non-spherical crystals, mathematical corrections are required to convert CLD data into CSD, which remains a significant challenge. Despite ongoing research to refine these conversion algorithms, this limitation remains a drawback of FBRM technology (Zhang et al., 2024). Although FBRM is well established for monitoring small molecule crystallization, its application to biological macromolecules like proteins remains limited, with only a few publications exploring its potential in this area (Walla et al., 2023). Combining PVM with FBRM enables the differentiation of precipitate identity (De Anda et al., 2005). For example, PVM can be used to differentiate bubbles, oils, and protein crystals that will contribute to the CLD regardless of their identity. PVM also provides turbidity information that can pair well with chord length distribution to fully describe the stages of a precipitation process (De Anda et al., 2005).

1.2. Modeling approaches

Modeling plays a crucial role in understanding the fundamentals of a system (Pistikopoulos et al., 2021; Mansouri et al., 2025). While empirical data helps detail observations from the phenomena, models are used to approximate and predict system behaviors that are difficult to measure. Validated models, by simulating processes, help reduce costs by minimizing the need for extensive and time-consuming physical experimentation. Utilizing models can enhance quality control by helping to understand the relationship between process variables and product quality, enabling better control over manufacturing processes. Certain fields, such as chemical engineering, rely on fundamental principles like conservation laws, contrasting with the widespread use of data-driven or black-box modeling in areas like image recognition and natural language processing (Chakraborty et al., 2022). A purely black-box approach may struggle to generalize to unseen data and risks overfitting, while a strictly mechanistic model faces challenges due to the many variables and parameters that are difficult to measure or estimate (Chakraborty et al., 2022). Therefore, there is an interest in research to develop data-driven models and combining them with first principles/mechanistic knowledge.

The combination of mechanistic models and machine learning (ML) models is pivotal in protein precipitation, merging mechanistic models for the phenomenology, and data-driven methods to enhance prediction accuracy and parameter estimation. This integration is crucial in process and product engineering, providing adaptable, reusable, and predictive frameworks for addressing complex challenges (Mansouri et al., 2025). In pharmaceutical production, where process optimization and shortened development cycles are paramount, this hybrid modeling approach offers numerous advantages. The ability of this approach to assimilate various knowledge sources, including prior information and

data-driven insights, makes it adaptable to diverse processes and complexities. Leveraging these models combining first-principles/mechanistic knowledge with data from experiments in protein precipitation facilitates parameter optimization and enables the prediction of particle size distribution and agglomeration dynamics during precipitation. This results in increased process efficiency, reduced costs, and improved product quality by tackling operational challenges, optimizing parameters, and making informed decisions based on system understanding and data-driven insights. For more information around this topic, we guide the readers to the following papers, Schubert et al. (1994) and Zhao et al. (2023) for a detailed explanation of using hybrid modeling, particularly in yeast fermentation processes. We would also like to draw attention to Hamilton et al. (2017) that developed hybrid models for predictions of a class of dynamical systems, and Nielsen et al. (2020a), Nielsen et al. (2020b) that used a hybrid model predictive control strategy for particle processes such as crystallization, flocculation or precipitation.

Combination of mechanistic models and ML models have been utilized in flocculation and related processes to predict kinetics and process parameters (Nielsen et al., 2020a; Nazemzadeh et al., 2021), assist in optimization, and aid in the development of soft sensors and process control (Nielsen et al., 2020a; Boskabadi et al., 2025, 2024). Several structural approaches exist in the literature for combining mechanistic models and ML models, either in parallel or in series, to improve process modeling.

In this work we have adapted a systems approach to model protein precipitation using lysozyme as a model product. The approach is capable of being applied to similar protein systems, both in the population balance modeling (PBM) (to estimate the particle size distribution from the PBM parameters) and in the surrogate modeling/ML methods (to correlate the PBM parameters to process operating conditions) to provide a more comprehensive understanding of the precipitation dynamics of batch proteins (Nielsen et al., 2020a). Furthermore, Samad et al. (2011) developed a generic multi-dimensional model for batch cooling crystallization, and Meisler et al. (2014) modeled crystallization kinetics with a generic modeling framework.

The manuscript is organized as follows: Section 2 presents the modeling framework with details on the population balance model and empirical correlations that estimate the particle-size distributions from operating parameters. The details of the surrogate model (using ML) to correlate the PBM parameters against the process operating conditions for fast estimation of PBM parameters, are also mentioned in this section. Section 3 provides details of the experimental procedure and data acquired from the process, using the FBRM probe as well as the experimental setup. Section 4 presents results from the application of the modeling framework to the case of precipitation of lysozyme (as described in Section 3). It also shows how the framework is used for the estimation and prediction for the particle-size distribution, sensitivity with changing PBM parameters, and predictions of the PBM parameters from the process operating conditions via a surrogate model. We also showcase in this investigation the limitations to further future research.

2. Modeling framework

The goal of this work is to obtain a dynamic model to predict precipitation of a specific product at different process operating conditions. With limitations on the exact know-how to aid in this estimation of the process dynamics, we have envisioned a framework that combines the population balance model (PBM) for protein precipitation, with a surrogate model for estimation of critical parameters in the PBM. Towards this goal, first, the PBM parameters are regressed based on each experimental dataset of particle size distribution, and subsequently, the surrogate model is trained to predict the regressed PBM parameters as a function of the process operating conditions. The surrogate model can then be used to predict the PBM parameters directly from the operating conditions, which can then be used to predict the particle size distribution (PSD) using the PBM model. Fig. 2 shows the detailed overview of the modeling framework. In the following sections these steps are explained in detail.

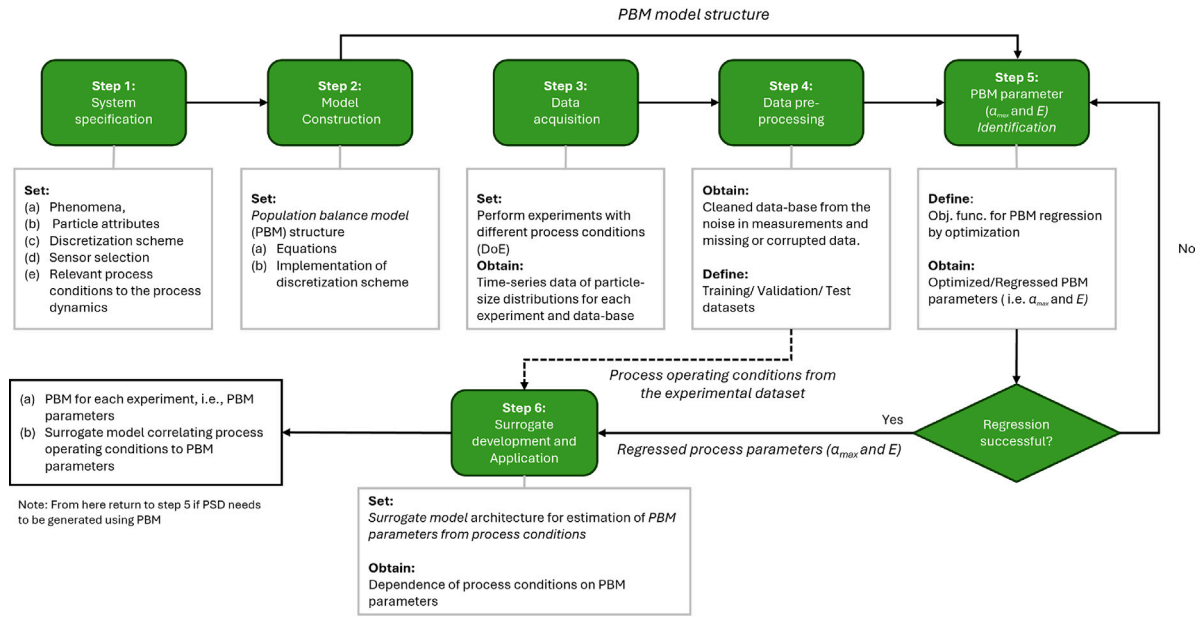


Fig. 2. Modeling framework overview.

2.1. Step 1: System specification

To design an efficient protein precipitation process, it is essential to understand the system and develop predictive models that can save both time and costs. We first chose a known and robust protein precipitation system to generate data for modeling. Then, a mechanistic model is employed to provide accurate predictions of the system's behavior.

In this work, a one-dimensional PBM is used as the mechanistic model to predict the dynamics of the precipitation process. Different phenomena and mechanisms can lead to the birth and death of particles. These phenomena include agglomeration, growth, nucleation, shrinkage and breakage, etc. The PBM contains parameters that will be discussed later and are either difficult to measure or estimate. ML methods can be used to assist in estimating these parameters. It is assumed in this study that only agglomeration and breakage phenomena are present, the effects of nucleation and growth are neglected. At the start of the experiment, all particles are assumed to be primary particles, as lysozyme crystals are initially seeded into the solution, thus the assumption of no nucleation. Multiple coefficients are employed within the equations to characterize the interrelations within the system. The PBM considered in this study is one-dimensional, describing the evolution of the mean chord length of protein precipitates as measured by FBRM. The model includes agglomeration and breakage phenomena in a single dimension, as this is sufficient to capture the primary dynamics of the system without introducing unnecessary complexity. A one-dimensional PBM is justified because the objective is to model the dynamic changes in chord length distribution rather than tracking the full particle size distribution. While a higher-dimensional PBM could provide more detailed insights, the available measurement technique (FBRM) primarily provides chord length distributions, making a 1D representation the most practical choice.

2.2. Step 2: Model construction

The model structure is envisioned as a two module system, that (1) estimates the particle-size distribution given the PBM parameters, and (2) correlates the process operating conditions with PBM parameters using a surrogate model.

$$\frac{\partial n_i(v, t)}{\partial t} = B_i - D_i \quad (1)$$

Eq. (1) denotes a generalized population balance model, with terms B_i and D_i corresponding to the birth-rate and death-rate of the particles in the i th size bin. Fig. 3 is a good visual representation of this phenomenon of birth and death where the highlighted particle in black is a representative particle of the i th particle size bin. This equation in the continuum limit and phenomenological interactions results in Eq. (2).

$$\begin{aligned} \frac{\partial n(v, t)}{\partial t} = & \frac{1}{2} \int_0^v \alpha(v-u, u) \beta(v-u, u) n(v-u, t) n(u, t) du \\ & - \int_0^\infty \alpha(v, v') \beta(v, v') n(v, t) n(v', t) dv' \\ & + \int_v^\infty S(u) \gamma(v, u) n(u, t) du \\ & - S(v) n(v, t) \end{aligned} \quad (2)$$

where n is the number density function of particles, t is time, v and u are agglomerate or particle volumes, α is the collision efficiency of particles, β is the collision frequency, S is the breakage rate, and γ represents the breakage distribution of particles. The first two terms are for agglomeration phenomena and the second and fourth terms account for the death of particles (Nazemzadeh, 2022).

Eq. (1) can be discretized into a general Eq. (3) in which the number of particles associated with a characteristic length over time is explained by birth and death terms.

$$\frac{dN_i}{dt} = B_i - D_i \quad (3)$$

Based on the assumptions made and the discretized terms in Table S1 (provided in the supplementary material) for birth and death terms for each present phenomenon, the discretized number of particles associated with each characteristic length over time can be written as Eq. (4).

$$\begin{aligned} \frac{dN_i}{dt} = & \left(\sum_{k=0}^m \sum_{j=k}^m (1 - 0.5 \cdot \delta_{j,k}) \alpha_{j,k} \beta_{j,k} \eta_{j,k,i} N_j N_k \right) \\ & - N_i \sum_k \alpha_{i,k} \beta_{i,k} N_k \\ & + \left(\sum_j \gamma_{j,i} S_j N_j \right) \\ & - S_i N_i \end{aligned} \quad (4)$$

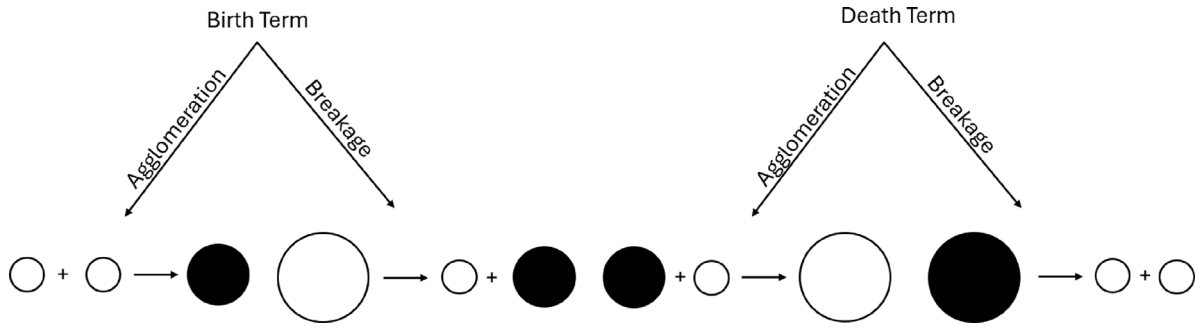


Fig. 3. Different phenomena and their contribution to both birth and death terms of the population balance model — black particle represents the agglomerate of class i .

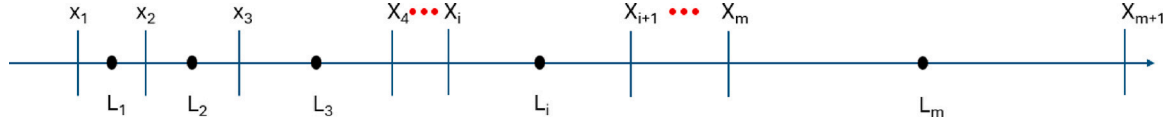


Fig. 4. A Schema of particle length discretization.

where, N is the discretized number density function, v is the characteristic volume of the particle bins, δ is the Kronecker delta, α is the collision efficiency, β is the collision frequency, η is the proportionality coefficient, γ is the breakage distribution function and S is the breakage rate. Details of the correlations and their discretized form are shown in the following paragraphs. The PBM parameters which can be tuned to specific experiments are $\alpha_{\max}$ and E as shown in Eqs. (8) and (9).

The particle size domain is discretized by the method of lines. The size domain is subdivided into m bins. The discretization is chosen to be a geometric series to account for larger particle sizes and to keep the computational load lighter. Eqs. (5), (6), and (7) are used to build the geometric grid of the discretization scheme (see Fig. 4).

$$R = \frac{L_i}{L_{i-1}} - 1, \quad (5)$$

$$L_i = (1 + R)L_{i-1},$$

$$R = \left(\frac{L_{\max}}{L_{\min}} \right)^{\frac{1}{i-m}} - 1, \quad (6)$$

$$x_i = \frac{2L_i}{(2 + R)}$$

$$\Delta L_i = 2\Delta_i = \frac{2RL_i}{(2 + R)} \quad (7)$$

Δ_i is the difference between the bins' characteristic length with the edges of the bins. The bins then become from $[L_i - \Delta_i, L_i + \Delta_i]$. In this case, R must be in the range $0 < R < 1$ and also the $L_{\min} = 2 \text{ nm}$ and $L_{\max} = 1 \text{ cm}$. The increasing size of the bins is typically used when a large range of particle sizes should be covered. Under exponential distribution, the number of particles of large sizes decreases approximately in a geometric progression. Thus, the bins are selected in such a way that initially, they should contain approximately equal particle numbers. It is assumed that all particles assigned to bin i have the representative length L_i .

2.2.1. Population balance model: Phenomenology and key parameters

In a system with agglomeration and breakage, phenomenologies that need to be considered include collision efficiency and breakage rate. In this section we explain these key phenomenologies in conjunction with adequate empirical models for additional parameters in the population balance model. There is a lack of library for these additional correlations, but we have considered these correlations to account for the phenomenology.

Collision Efficiency: The collision efficiency is the probability of two flocculates (aggregated particles formed through flocculation,

where suspended particles in a liquid bind together due to inter-molecular forces or the addition of coagulants) staying together after colliding. It is generally a value between [0-1] or a function of the sizes of the two colliding particles. In simpler terms, the parameter represents the fractions of successful collisions in the agglomeration process. The function assumes that the sticking probability between any two particles is related to the difference in their speed, which depends on their sizes, with all that the function used by Li et al. (2019), has been used in this study.

$$\alpha_{i,j} = \alpha_{\max} \left(\frac{L_i/L_j}{1 + L_i/L_j} \right)^2 \quad (8)$$

$\alpha_{\max}$ is considered one of the parameters of the PBM to be changed to fit the model into experimental data. It should be noted that Eq. (8) produces a non-symmetric matrix, this means that the collision efficiency between particle i and particle j is different from the collision efficiency of particle j and particle i , to address the issue it is assumed that $\alpha_{i,j} = \alpha_{j,i}$ which is that the collision efficiency would then be much simpler and symmetrical.

Breakage rate: The term S in Eq. (2) represents the breakage rate of agglomerates to smaller particles. The breakage mainly takes place because of the induced shear made by the mixer. It is assumed that the breakage rate is always proportional to the size of the particles (Nazemzadeh et al., 2021). There are many ways to determine the breakage rate, but the simplest one that is used in this study is proposed by Li et al. (2019).

$$S_i = E \left(\frac{L_i}{L_p} \right)^{\frac{2}{D_f}} L_p \quad (9)$$

where E is the breakage rate constant or breakage rate, coefficient, related to the induced shear in the system. The fractal dimension has been assumed to be 2 for simplification. This assumption was based on the observation that lysozyme, although inherently asymmetric, is generally observed to form a roughly spherical shape upon precipitation (Andersson and Hansson, 2018). It is acknowledged that while this simplification serves as a useful starting point, future work may explore variations in the fractal dimension to capture more nuanced effects on the crystal structure.

Collision frequency: The collision frequency is correlated with the motion of particles in the bulk medium. The pattern of particle movements can play an important role in the estimation of a function for the collision frequency. Three main mechanisms exist for agglomeration processes (Nazemzadeh, 2022):

- **Perikinetic agglomeration:** This process involves the agglomeration of particles due to their random motion, known as Brownian motion. Brownian motion is characterized by the random and chaotic movement of particles or molecules, which is highly influenced by the system's temperature. It becomes notably slower for particles larger than 1 μm and can generally be disregarded for larger particles.
- **Differential sedimentation:** The motion of particles in vertical and in the direction of the bottom of the vessel. The settling velocities of particles would be different based on the difference in their size and densities.
- **Orthokinetic agglomeration:** Also known as shear-induced agglomeration, this mechanism occurs due to the movement of particles induced by velocity gradients within the system, typically generated by impellers inside the Vessel. It finds extensive applications across various industrial processes. The shear and mixing created by these velocity gradients lead to a higher frequency of collisions between particles. Additionally, it is important to note that increased shear can also enhance the breakage of weakly attached particles.

$$\beta_{\text{global}} = e_{\text{Br}}\beta_{\text{perikinetic}} + e_{\text{DS}}\beta_{\text{DS}} + e_{\text{Sh}}\beta_{\text{orthokinetic}} \quad (10)$$

where β_{global} is the global frequency, $\beta_{\text{perikinetic}}$ represents the frequency due to the Brownian motion effect, β_{DS} is the frequency corresponding to differential sedimentation, and $\beta_{\text{orthokinetic}}$ is the frequency caused by the induced shear in the system.

There are two approaches to estimate the collision frequency, the rectilinear approach which assumes that the liquid medium does not influence the particle trajectories and that the two colliding particles follow a rectilinear line. The other approach is called curvilinear and assumes that when two particles approach each other, the liquid film is squeezed out, and it causes rotations and changes in trajectories of the particles to be curvilinear, this means a lower collision probability while the rectilinear approach overestimates. In the global frequency equation for a rectilinear approach, all the weights for the frequencies will be 1 and equal, while for a curvilinear approach, they will have a value between [0-1].

Since the majority of the particles measured by FBRM tend to have a trend of being near 5–20 μm , based on the research done on agglomeration (Nazemzadeh, 2022) even though they have acknowledged the importance of sedimentation agglomeration for medium-sized (between 1 to 100 μm) particles, they have neglected the significance of other mechanisms and also assumed a rectilinear approach for further simplification.

Given the presence of shear-induced mixing in the vessel and the detection of many particles above 1 μm by the Focused Beam Reflectance Measurement (FBRM), to simplify the system only the shear-induced agglomeration has been assumed. The collision frequency due to the shear rate of the mixing vessel is considered to be a function of particle size. The correlation for β is the following. G is the average velocity gradient and is related to the mixer's rotation speed and the viscosity of the system.

In the literature, the orthokinetic agglomeration collision frequency has been found for two colloidal particles of different sizes (Nazemzadeh, 2022):

$$\beta_{i,j} = \frac{G}{6} (L_i + L_j)^3 \quad (11)$$

The correlation, where G represents the mean velocity gradient and L denotes the particle size and has been widely utilized for agglomeration studies. However, a key assumption underlying its application is that aggregates should ideally be spherical. In scenarios involving agglomerates with irregular shapes, formed from the agglomeration of very small particles into fractals, a modified frequency expression is

introduced. This modification incorporates the irregular shapes characterized by fractal dimensions (Nazemzadeh, 2022; Li et al., 2019).

$$L_c(i) = \left(\frac{L_i}{L_p} \right)^{\frac{2}{D_f}} L_p \quad (12)$$

where L_p is the primary particle size, which for lysozyme protein is around 2 nm and D_f is the fractal dimension.

$$\beta_{i,j} = \frac{G}{6} (L_c(i) + L_c(j))^3 \quad (13)$$

The average velocity gradient can be calculated from the following correlation (Li et al., 2019)

$$G = \sqrt{\frac{\varepsilon}{\nu}} \quad (14)$$

where ε represents the homogeneous turbulent energy dissipation rate of the mixed tank and ν is the kinematic viscosity of the suspending fluid.

An estimation for the maximum turbulent energy dissipation rate of the mixed tank can be derived from the following empirical correlation (Peter et al., 2006):

$$\varepsilon_{\text{max}} = 0.1 \frac{\omega^3}{h_1} \quad (15)$$

where $\omega = \pi N D$ is the angular velocity of the stirrer. N is the frequency of the stirrer (rpm) and D is the diameter of the stirrer (m), h_1 is the height of the stirrer (m).

Then from Eq. (16) the average energy dissipation rate for unbaffled flasks may be calculated (Peter et al., 2006).

$$\varepsilon = N e' \cdot N^3 \cdot d^4 \cdot V_L^{-\frac{2}{3}} \quad (16)$$

where V_L represents the filling volume of the flask, and $N e'$ is the modified Newton number (power number) of the shake flask and d is the inner diameter of the container.

Modified Newton number can be calculated from the following equation (Peter et al., 2006):

$$N e' = 75 Re^{-1} + 25 Re^{-0.6} + 1.5 Re^{-0.2} \quad (17)$$

where the Reynolds number is defined as:

$$Re = \frac{\rho_c \cdot N \cdot d^2}{\eta} \quad (18)$$

where ρ_c is the density of the solution, N is the frequency of the rotation, d is the inner diameter of the container and η is the dynamic viscosity of the solution.

Proportionality coefficient: The proportionality coefficient(η) determines the fraction of newly formed particles to the size class or bin. This is crucial to the discretized PBM to ensure the preservation of mass and number balance.

$$\eta_{j,k,i} = \begin{cases} \frac{v_{i+1} - (v_j + v_k)}{v_{i+1} - v_i}, & v_i \leq (v_j + v_k) \leq v_{i+1} \\ \frac{(v_j + v_k) - v_{i-1}}{v_i - v_{i-1}}, & v_{i-1} \leq (v_j + v_k) \leq v_i \end{cases} \quad (19)$$

v_j and v_k are the two particles that are colliding. The result of this collision then would have a volume of $v_j + v_k$. Since the resulting particle might not directly be the characteristic bin volumes. It will fall between v_{i-1} , v_i and v_{i+1} in the discretization. Since The PBM solves the number of particles for the characteristic volumes or lengths, it is important to assign the newly made particles to their respective bins. The proportionality coefficient ensures that a portion of the new particle goes to the closest bins near that new particle. There is a special case as well where the newly formed particle volume is larger than the last bin's characteristic volume. In that case, the formula will change into the following:

$$\eta_{j,k,m} = \begin{cases} \frac{(v_j + v_k)}{v_m}, & v_m \leq (v_j + v_k) \end{cases} \quad (20)$$

Breakage distribution: Apart from the breakage rate Eq. (9), the breakage distribution can be defined to explain the distribution of smaller particles from breakage of an agglomerate (Li et al., 2019; Nazemzadeh et al., 2021). Similar to the proportionality coefficient defined for agglomeration, since in the breakage phenomena, it is assumed that the total mass of particles and the volume of particles, which is a simpler representation of mass, will be constant. The assumption is that in the breakage no new mass is added or removed from the system, so the total mass of the particles and directly the total volume of the particles would be constant. The breakage distribution function (γ) of an agglomerate determines the size of daughter particles resulting from breakage. Two common approaches are the simple binary breakage function and the normal breakage function (Li et al., 2019; Nazemzadeh et al., 2021). Previous studies suggest that daughter particle generation follows a normal distribution (Li et al., 2019; Nazemzadeh et al., 2021). For a linear discretization grid, this can be approximated by a binomial distribution function.

$$\gamma_{j,i} = \frac{\binom{j}{i}}{\sum_{j \geq i} \binom{j}{i}} \quad (21)$$

In Eq. (21), γ represents the distribution function that describes how a particle of volume v_j breaks into smaller particles. This function is based on the probabilities or possible pathways through which the particle can fragment. To simplify and enhance the understanding of the equation, a third variable k can be introduced. The equation is then rewritten as follows:

$$\gamma_{j,i} = \frac{\binom{j}{i}}{\sum_{k=i}^j \binom{k}{i}} \quad (22)$$

In the case of a linear grid, the distribution is sufficient and the mass balance is conserved. However, this is not the case for a geometric grid, and adjustments based on the volume of particles are necessary. This can be achieved by normalizing the function with the volume of particles as weights. To ensure a constant or nearly constant total particle volume, several equations were tested.

$$\gamma_{j,i,modified} = \frac{\gamma_{j,i} v_i}{\sum_{k=i}^j \gamma_{k,i} v_k} \quad (23)$$

Note that there are two constraints for $\gamma_{j,i}$. The first is that all the volume fractions distributed to the size bin i should add up to make the volume of bin i for a given breakage of a particle size j .

$$\sum_{k=i}^j \gamma_{k,i} v_k = v_i \quad (24)$$

$$\gamma_{j,i} = 0 \quad \text{for } k > i \quad (25)$$

The second constraint dictates that for instances in which $i > j$, the fractions must be nullified, as it is inconceivable for a larger particle to emerge from the breakage of a smaller particle.

2.3. Step 3: Data acquisition

The data acquired in this work are the process operating conditions and the output chord-length distribution (also referred to as PSD), for each operating condition. The chord-length distribution is usually collected with a given frequency which needs to be provided to the model, during the optimization state of the problem.

Data for chord-length distribution is observed as the number of particles of a particular size as seen by a sensor of choice. These are often paired as $(\{L_i, N_{i,t}\})$ where L_i is the chord-length and $N_{i,t}$ is the number of particles recorded by the sensor at time t , for chord-length L_i . The resultant data acquired is a multidimensional time-series.

2.4. Step 4: Data preprocessing

The data acquired is converted to a chord-length distribution based on the discretized scheme, this means that the measurements are put into their respective particle size bins made in the discretization step to ensure the same resolution with the modeling results.

These distributions are then normalized to preserve the dynamic characteristics of the system and ensure consistency across different experiments and simulation runs. The normalization step is crucial, as it aligns the scale of experimental observations with that of the model, particularly in terms of particle population, making the comparison between the two meaningful and accurate. Additionally, this transformation enables interpolation across chord-length ranges that may not have been directly captured during the measurements, allowing for a more continuous and representative distribution to be used in model evaluation.

To ensure numerical stability and robustness during the simulation and optimization, further preprocessing is applied. Negative values in the simulated outputs are removed to maintain physical realism, and a minimum threshold is imposed to prevent division by zero during subsequent calculations. Both simulated and experimental distributions are normalized into probability distributions by dividing each by their total particle count.

2.5. Step 5: PBM parameter (α_{max} and E) identification

In this study, we consider agglomeration and breakage to be the primary phenomena governing the precipitation process. A PBM was developed based on these phenomena to predict the system's dynamics over time. The PBM predicts particle length distributions for each experiment, with model parameters influencing the kinetics of precipitation.

For each experimental dataset, the counts of measured chord lengths are redistributed into the corresponding PBM bins by summing the contributions of data points that lie within each bin's boundaries. This ensures that the measured distributions are represented at the same resolution as the PBM, allowing for direct comparison and integration into the modeling framework.

The objective function for the optimization has to represent the similarity between the predictions and the measurements. Although several objective functions can be defined and would all work, using a simpler and more efficient approach enhances computational speed. Therefore, instead of complicating the objective function, the KL divergence between predicted values and measured values has been chosen as the objective function. The use of KL divergence in the context of protein size distributions has already been reported in the literature (Maddux et al., 2017).

Kullback–Leibler (KL) divergence is a measure of how one distribution diverges from a second, reference distribution. It quantifies the difference between the simulation distribution P_i and measured distribution Q_i . Mathematically, KL divergence is not symmetric, meaning $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$, and it is always non-negative. The formula for KL divergence and the objective function are presented in Eqs. (26) and (27). KL divergence between the simulated particle size distributions over time and measured particle size distribution over time has been chosen as the objective function.

$$D_{KL}(P \parallel Q) = \sum_i P_i \log \left(\frac{P_i}{Q_i} \right) \quad (26)$$

$$\begin{aligned} \text{Objective function} &= \min_{\{\alpha_{max}, E\}} \sum_t D_{KL}(P \parallel Q) \\ &= \min_{\{\alpha_{max}, E\}} \sum_t \sum_i P_{i,t} \log \left(\frac{P_{i,t}}{Q_{i,t}} \right) \end{aligned} \quad (27)$$

The simulation distribution $P_{i,t}$ and measured distribution $Q_{i,t}$ are the normalized values of the solution of the PBM and the measured

particle distributions, respectively. Since KL divergence only takes the distribution that sums to unity, the distribution has to be normalized. These normalized distributions are calculated using Eqs. (28) and (29).

$$P_{i,t} = \frac{N_{i,t}}{\sum_j N_{j,t}} \quad (28)$$

$$Q_{i,t} = \frac{N_{measured,(i,t)}}{\sum_j N_{measured,(j,t)}} \quad (29)$$

The objective function has been defined as the sum of the KL divergences of the distributions for every 30 min. After this step, we obtain optimized α_{max} and E for each experiment. However, the compute required for this step is rather expensive and time-consuming. A surrogate model for estimation of PBM parameters from the process operating conditions, is helpful hence to correlate the process operating conditions with optimized PBM parameters.

2.6. Step 6: Surrogate development and application

A key problem to be addressed by the framework is to correlate the PBM model parameters, specifically α_{max} and E , with the process operating conditions. To correlate the process operating conditions to the parameters of the PBM, a surrogate model is required that can adequately estimate these PBM parameters.

In our case, an artificial neural network-structure with 1 input-layer with 4 inputs (corresponding to the process operating conditions), 1-output layer with 2 outputs (corresponding to α_{max} and E), and 2 hidden layers with 64 neurons and 32 neurons, respectively, was used. The model has in total 3 layers apart from the input layer, with a ReLU activation function after each layer. Implementing dropout layers helps the model by reducing over-fitting. The neural network architecture was built using the PyTorch module on Python. Supplementary material (Table S4) provides more details about the neural network structure. Details on training are elucidated in Step 6: Surrogate development and application.

The surrogate model is structured to take the process operating conditions, such as protein concentration, salt concentration, rotation speed, and temperature for each experiment as inputs, to predict the PBM parameters (α_{max} and E) for each experiment. Given the computational cost to predict the PBM parameters for each individual experiment, and the experimental effort to collect the process operating conditions data alongside the PSD, a surrogate model is required to correlate the process operating conditions to PBM parameters, for better design of experiments. This trained model can be used to predict the optimized α_{max} and E , obtained after step 5. To correlate the process operating conditions to the optimized parameters of the PBM, any ML model may be used.

Mean Squared Error (MSE) loss function and Adam optimizer are used to train the surrogate model that correlates the process operating conditions to PBM parameters. The MSE loss calculates the average of the squares of the differences between the predicted and actual PBM parameter values (which are obtained post-optimization in step 5), inherently penalizing larger errors more significantly; this quality makes it especially suited for regression tasks where precise numerical predictions are crucial.

Since the number of samples is minimal, it was decided to use cross-validation and training to improve the regularization of the model. The model was trained on 10,000 epochs to ensure sufficient learning. The neural network's loss optimizer used mean squared error between the predictions and the training values as the objective function. More information is given in the supplementary material (Section S6). At the end of step 6, we have successfully obtained PBM parameters, i.e., the regressed PBM parameters for each experiment. We have a surrogate model that can predict the regressed process parameters from the process operating conditions.

3. Lysozyme precipitation: Experimental setup

We chose a simple and robust protein–buffer system of lysozyme in sodium acetate pH 4.0 to demonstrate the usefulness of process analytical technologies. At low pH, lysozyme is highly charged and will not interact with itself in solution. The addition of salts like sodium chloride influences precipitation by modifying interactions, with the rate depending on solution molarity. This well-characterized crystallization system of lysozyme in sodium acetate pH 4.1 buffer was chosen as a starting point for developing a robust precipitation system which preserves functional proteins (Judge et al., 1999). Lysozyme has a pI value of 10.7, and is therefore expected to be fully charged at pH 4.1 (Baron et al., 2016). The published lysozyme sodium acetate with NaCl process produced tetragonal crystals without shear, suggesting that the interaction kinetics were amenable to a controllable precipitation process. Shear was added to test precipitation kinetics at various protein, salt, temperature, and agitation rate combinations.

The 0.1 M sodium acetate pH 4.1 solution was made in deionized water and adjusted to pH 4.1 with glacial acetic acid. The reported buffering capacity of sodium acetate is pH 3.6 to 5.6. Lysozyme and NaCl were combined in the base sodium acetate buffer. All solutions were filtered through a 0.1 μ m filter prior to vessel addition.

Precipitation screening was achieved by adding increasing amounts of salt and protein and reacting overnight at 25 °C at a moderate shaking speed in a 96-well format (cat. 3788, Corning) and using binary well scoring for visible precipitation.

3.1. Process analytical technology

Dynamic light scattering

To monitor sub-visible particle–particle interactions, dynamic light scattering (DLS) with the DynaPro Plate Reader III™ (Wyatt Technology) was used to identify the appropriate buffer range and determine aggregation propensity in solution at low to modest NaCl concentration in 96-well batch format (cat. 3603, Corning).

Process analytical technology

To monitor the precipitation process, a Mettler Toledo EasyMax 102 workstation with a 100 mL vessel was setup with a five-port headplate. The EasyMax workstation provides temperature and agitation process control and syncs with the process monitoring probes. The stirring crossbar configuration was chosen to accommodate the necessary probes. The focused beam reflectance measurement (FBRM) G400 probe and the EasyViewer 100 probes were submerged in the vessel. The FBRM measures CLD and particle counts. The EasyViewer monitors turbidity and records particle images. Data collection from both the FBRM and the EasyViewer was done in the iControl software.

The probes were submerged in the lysozyme only solution to establish a baseline. The NaCl solution was added ten minutes after lysozyme and the entire precipitation was monitored for 24 h.

EasyMax 102

The experiments in this study were conducted on an advanced synthesis workstation, EasyMax 102, from Mettler Toledo. A 100 mL glass vessel attached to a temperature probe, FBRM probe, Turbidimeter, and magnetic stir bar is utilized as shown in Fig. 5. A deep vortex is created by the use of a magnetic stir bar. Temperature is maintained at a set point by a water-cooled jacket. An overview of the experimental setup is provided in the supplementary material (figure S1).

ParticleTrack G400

A FBRM (focused beam reflectance measurements) model G400 Mettler-Toledo probe was used for this study. The instrument ParticleTrack G400 uses an FBRM probe-based technique. A rotating monochromatic laser beam is used to analyze backscattering, and FBRM can be used even in concentrated slurries. A particle's optical properties play a crucial role in FBRM since it is based on scattered laser light reflection on encountered particles in the system. The instrument is inserted at an angle into the process stream to ensure

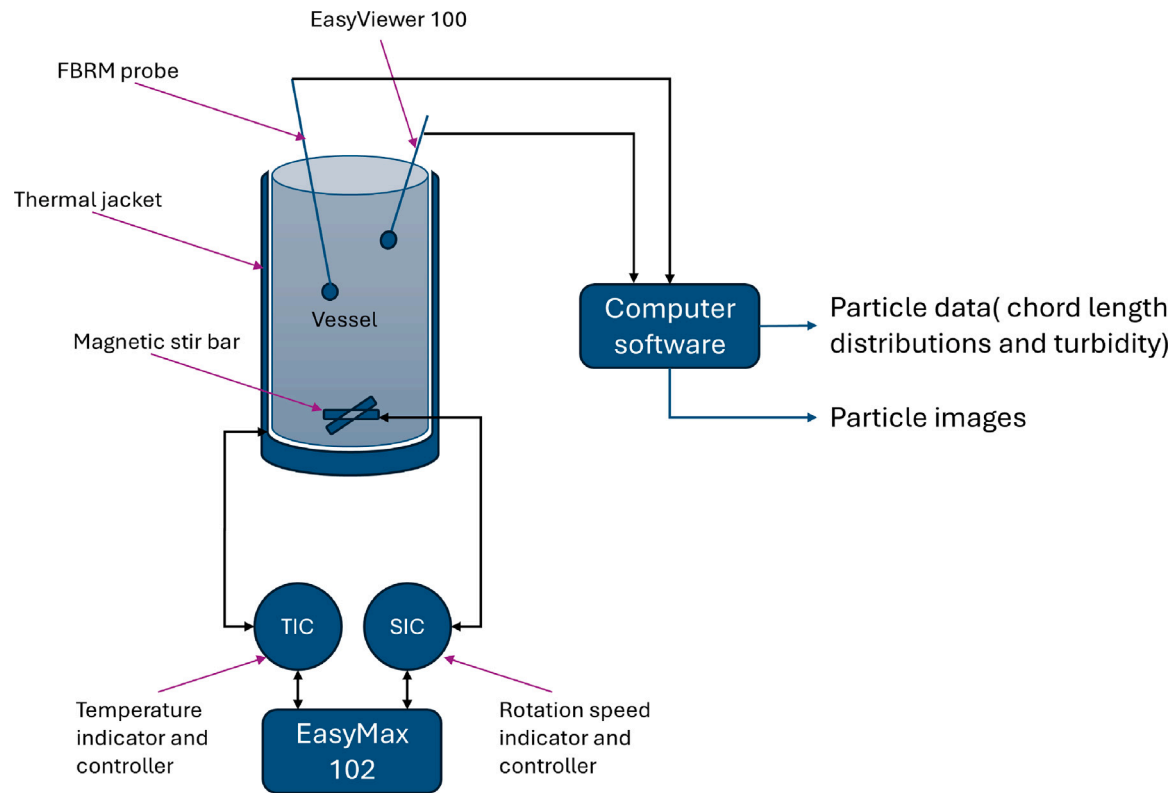


Fig. 5. A block flow diagram of the experimental setup.

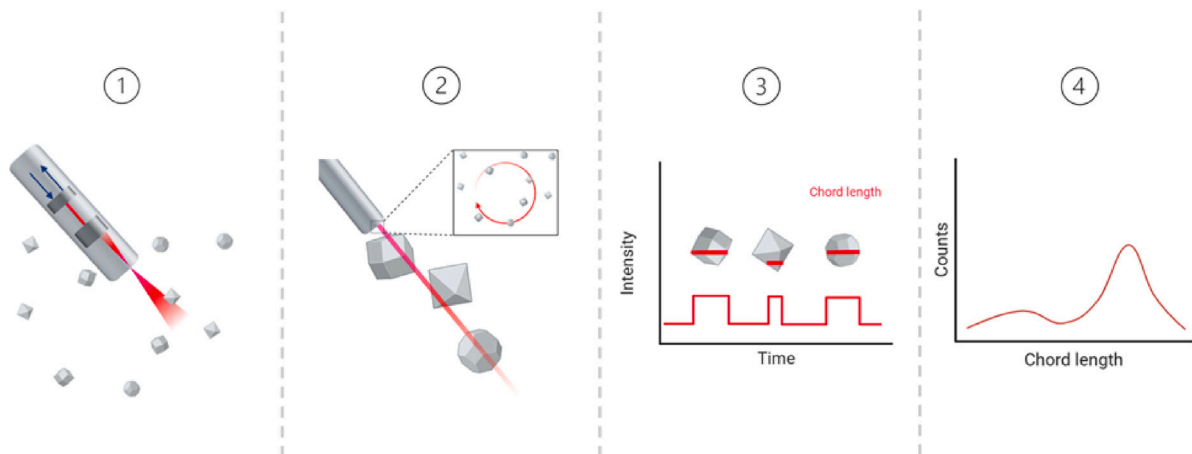


Fig. 6. Conducting FBRM measurements by ParticleTrack G400 for particle characterization. (1) Immersion of FBRM probe into solution. (2) Scattering occurs when laser beams detect particles in the sapphire windows. (3) Scattering light intensity is used to collect data. (4) The chord length distribution is determined.

proper particle flow across the probe (Walla et al., 2023). Samples are not required to be collected or prepared. The FBRM beam measurement with ParticleTrack G400 is shown in Fig. 6.

To focus a laser beam on a narrow spot at the sapphire window, a set of optics is placed down the probe tube. By rotating the optics at a fixed speed, typically 2 m/s, the beam spot rapidly scans across particles as they pass the window. The laser beam scans throughout the particle system, and as it reaches an individual particle or more, the laser light will be backscattered to the detector. When the focused beam reaches a particle, the laser light reflects to the detector. These pulses of backscattered laser light are counted, and the particle distance is determined by multiplying the duration of each pulse by scan speed. The particle distance known as the chord length is a fundamental measurement of particle size.

3.2. Data operating conditions

The time-series data of FBRM will be utilized to understand the kinetics of the lysozyme precipitation in the following modeling section. The data will be used for parameter estimation of PBM.

3.3. Data description and preprocessing

The data from FBRM is stored from the sensor into excel files in which there are counts corresponding to specific particle chord length. The data is noisy and especially at the beginning of the experiment, the bubbles that are formed during the initial start of the experiment interfere with the FBRM and result in an increase of measured particle chord length. The data needs to be cleaned from inconsistencies and

Table 1

An overview of the PBM model parameters that are known and unknown based on the assumptions in this study. PBM parameters are matrices. All the expressions for the parameters are provided in the following section. m is the number of discretization bins in this study.

Category	Parameter [Unit]	Description	Status	Dimension	References
Known	β [s^{-1}]	Collision frequency	Calculated from literature correlations	$[m \times m]$	Li et al. (2019), Nazemzadeh et al. (2021) and Peter et al. (2006)
	η [-]	Proportionality coefficient	Calculated from literature correlations	$[m \times m \times m]$	Li et al. (2019) and Nazemzadeh et al. (2021)
	γ [-]	Breakage distribution	Calculated from literature correlations	$[m \times m]$	Li et al. (2019) and Nazemzadeh et al. (2021)
Optimization variables	α [-]	Collision efficiency	To be optimized	$[m \times m]$	Li et al. (2019) and Nazemzadeh et al. (2021)
	S [s^{-1}]	Breakage rate	To be optimized	$[m]$	Li et al. (2019) and Nazemzadeh et al. (2021)
Solved	N [-]	Number of particles for each bin, if only the number of particles is solved	Solved from PBM equations	$[m]$	-

all the noises removed. Any corrupted or missing time points are also removed. The data is to be normalized to be used in the future steps of the modeling framework. The normalization is done to ensure comparability between the modeling particle size distributions and measured particle size distributions.

The measurement error of FBRM based on the technical documents provided by the manufacturer is 4% of the reference samples for the median of PSD for particles in the size range of one to 1000 μm . This is the allowed error specified by the manufacturer, however, the reported error from calibration validation is much less than 4%.

In the following section, we present the results and discussion for this application to lysozyme precipitation.

4. Application of the framework: Lysozyme precipitation

4.1. Step 1: System specification

The antimicrobial enzyme lysozyme is highly significant in the food and pharmaceutical industries (Nawaz et al., 2022; Ferraboschi et al., 2021). In food preservation, it inhibits bacterial growth, particularly Gram-positive bacteria, thereby extending the shelf life of products like cheese and wine (Nawaz et al., 2022; Ferraboschi et al., 2021). In the pharmaceutical sector, lysozyme is used as a model protein to study protein stability and inactivation, aiding in the development of stable formulations for therapeutic proteins (Ferraboschi et al., 2021). In this work, we are presenting a case study of the dynamics of batch precipitation of lysozyme. The results of PBM and measurements distributions and also the ANN model predictions are provided. The dynamics of lysozyme are modeled using a PBM, the parameters of the PBM are fine-tuned to the experimental data, and finally ML methods are used to correlate the effects of changing process operating conditions on the PBM parameters. Further details are given in section (S2) of the supplementary materials.

4.2. Step 2: Model construction

The PBM model introduced in Section 2.2 is implemented in Python, and the implementation codes are available upon request. In Table 1, an overview of the PBM is presented. Collision efficiency and breakage rate are being optimized in the outer optimization loop in this study. PBM parameters are matrices in which the number of elements is written in Table 1.

To ensure the developed model is solvable, a model analysis is carried out as described by Cameron and Gani (2011). Although at first glance the model seems to have a lot of variables apart from $N(t)$, which is the solution of the PBM, a more careful inspection reveals that the system consists of a discretized set of ordinary differential equations

Table 2

Incidence matrix for PBM.

PBM	$\alpha_{i,j}$	Re	Ne'	ϵ	G	$L_c(i)$	$\beta_{i,j}$	$\eta_{i,j,k}$	S_i	$\gamma_{i,j}$	N_i
Eq. (8)	x										
Eq. (18)		x									
Eq. (17)			x								
Eq. (16)				x							
Eq. (14)					x						
Eq. (12)						x					
Eq. (13)							x				
Eq. (19)								x			
Eq. (9)									x		
Eq. (23)										x	
Eq. (4)	x						x	x	x	x	x

along with ten algebraic equations (Eq. (4) and the associated constitutive equations, (8), (9), (12), (13), (14), (16), (17), (18), (19), (23) - see Incidence Matrix in Table 2) written in vector form. With m being the number of particle bins, the total number of variables becomes $N_{var} = 11m + 2$, consisting of m components for N , 10 m dependent algebraic variables calculated from their respective correlations, and 2 PBM parameters: α_{max} and E . The number of equations is $N_{eqs} = 11m$, resulting in a number of degrees of freedom $N_{dof} = 2$. These PBM parameters are specified in the outer loop through the optimizer to ensure that the system in the inner loop remains solvable. The incidence matrix for PBM is given in Table 2. It can be seen from Table 2 that it is a lower tridiagonal matrix which means the equations can be solved sequentially. Note, at time zero, t_0 , initial particle distribution, N_0 , is known. Given N_i plus model parameters and input conditions, $\frac{dN_i}{dt}$ is calculated. Given $\frac{dN_i}{dt}$, calculate for t_{new} and $N_{i,new}$. Repeat until $t_{new} = t_{end}$. For the optimization problem discussed in step 5, see section 2.5, N_i for all i is used to calculate the optimization objective function to compare with the measured data $N_{i,measured}$.

For the surrogate model, an artificial neural network structure with 2 hidden layers was used. The neural network was built using the PyTorch module on Python. Since the number of samples is minimal, it was decided to use cross-validation and training to improve the regularization of the model. Implementing dropout layers helps the model by reducing over-fitting. The model was trained on 10,000 epochs to ensure sufficient learning. Supplementary material (Table S4) provides more details about the neural network structure.

4.3. Step 3: Data acquisition

The data is acquired using the experimental setup described in Section 3. The experimental datasets are provided as supplementary materials.

Table 3

Overview of the experiments conducted to validate the modeling framework. The pH for all experiments is 4.1. A star (*) indicates that the rotation speed was set to 200 rpm, but the measured value is shown.

Experiment number	Protein concentration (g/l)	salt Concentration (M)	Rotation speed (rpm)	Temperature (°C)
1	3	1.7	200	20
2	2	1.7	200	20
3	5	1.5	200	20
4	2	1.5	198	20
5	10	1.5	200	20
6	2	1.5	199	20
7	5	2.0	300	20
8	5	2.0	100	30
9	10	2.0	100	20
10	5	2.0	300	30
11	10	2.0	300	30
12	10	1.0	100	20
13	10	1.0	300	20
14	10	1.0	300	30
15	5	2.0	100	20

Several experiments were conducted in the design space of four process operating conditions such as salt concentration, protein concentration, stirrer speed and temperature. The process operating conditions and experiment numbers are presented in Table 3. The FBRM probe measured the particle distribution and the data were collected.

4.4. Step 4–5: Data preprocessing and PBM parameter (α_{max} and E) identification

The acquired data in step 3 is processed according to the procedure explained in Section 2.4. The processed data is used to fine-tune PBM parameters based on the experimental measurement results and the simulation results. The results of the PBM modeling and optimization are discussed in the following sections.

The PBM model takes both the known and unknown parameters from Table 1, along with any system information provided in the supplementary material (Table S2), and calculates the number of particles for each bin over a specified time. Since we have unknown PBM parameters, to make the PBM solvable, an outer optimization loop is designed to fine-tune PBM model parameters such as α_{max} and E .

The experimental conditions, results, and optimized model parameters are summarized in Table 4. To ensure a fair comparison between datasets, only the experiments in which precipitation occurred in the first 24 h were chosen. To align the experimentally measured chord length distributions with the resolution of the PBM bins, a discretization approach was used. The PBM bins are defined by a central size and a range determined by their half-width. Each measured data point is assigned to a PBM bin if its value falls within the bin's range.

The differential evolution method from the `scipy.optimize` Python library was selected for optimization. While other methods were tested, local optimization algorithms often became trapped in local minima, failing to adequately explore the search space. The differential evolution method, although computationally intensive, running for nearly 7 h (2.2 GHz Intel(R) Core(TM) i9-13900HX), proved more effective.

The optimization parameters were selected to balance computational efficiency with the need for a thorough search, employing a high population size (popsize) and 3 iterations. To balance it out, a broader range for mutation and a higher value for recombination was defined to allow the optimizer to diversify. The seed number is used to ensure the reproducibility of the results. The optimizer's internal variables are given in detail in the supplementary material (Table S3).

4.4.1. Effects of changing population model parameters on the particle distribution

Based on the theory explained earlier, we predict that the distribution will lead towards indicating a greater agglomeration with the increasing of α_{max} , and will favor breakage with E increasing.

To understand the effects of model parameters on the particle distribution output from the PBM, the distribution was simulated while keeping one of the parameters constant and changing the other. A selection of 2 different conditions for each are plotted in Figs. 7 and 8. If the collision efficiency is set to zero, then nothing will happen, and if the breakage rate coefficient is set to zero, the system will just have agglomeration phenomena and will grow larger until it reaches a single particle. The jumps observed in the median predictions could be explained by the properties of the median as a measure of the center in the data. Unlike the mean, which continuously reflects small changes in all data points, the median is determined solely by the middle value of an ordered dataset. Consequently, minor fluctuations in the underlying distribution may not alter the median until a sufficient number of data points shift, leading to fast changes rather than smooth transitions. This characteristic means that even if the particles grow in size gradually, the median may only update once the central portion of the data crosses a threshold, thereby producing noticeable jumps in the prediction trend.

Increasing the breakage rate coefficient to a number above 3 will lead to mathematical instability in the model and it might be because of assumptions in the modified breakage distribution function which will lead to the creation of more particles than what is should be and it results in very high agglomeration at the end of the experiment. This is important to understand the range of values within which the PBM parameters can change, or useful for any optimizer that needs boundaries for the inputs.

4.4.2. Initial condition and system information

To balance computational efficiency with accuracy, the total number of particle bins (m) was set to 30. This binning choice ensures a sufficient resolution to capture particle distribution dynamics effectively. Any number above 30 will lead to significantly higher computational time. As can be seen in Fig. 9, although increasing the number of bins improves accuracy, using around 30 bins strikes a balance by being sufficiently close to larger values while significantly reducing computational time. The initial condition was constructed by assuming that all particles present, based on the specific protein concentration and reaction volume, act as primary particles in the experiment. Further details are provided in the supplementary material (Table S2). The ODE system was then solved using the `solve_ivp` function from Python's SciPy library, with the LSODA method chosen due to its robust handling of stiffness, which enhances solution stability and convergence. Further details about the solver are provided in the supplementary material (section 2.2).

4.4.3. Measured particle distribution and simulated particle distribution with the optimized model parameters from the population balance model

In Fig. 10, particle distributions of 4 different experiments with different regressed PBM parameters are plotted. The measurements

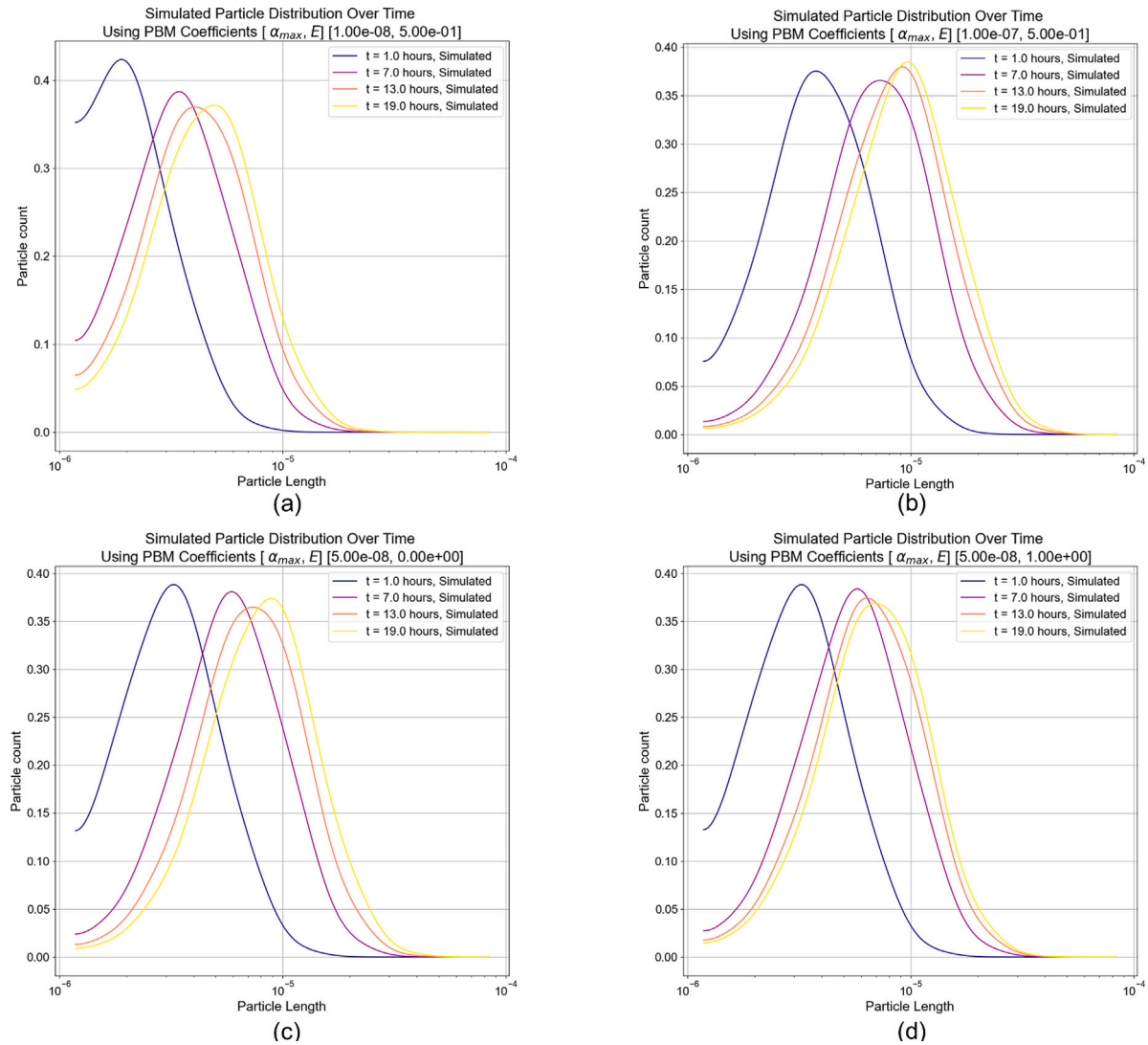


Fig. 7. Particle distribution overview and the effect of model parameters on the system, rotation speed for all of the models is considered 200 rpm. (a–b) Two cases in which the increase of efficiency is leading to the appearance of larger particles. (c–d) An increase in breakage coefficient will lead to smaller particles and also mathematical instability when the coefficient is near 3 or above.

Table 4

An overview of the experiments chosen to be used for PBM parameter optimization. The units for protein concentration (g/L), salt concentration (M), rotation speed (rpm), temperature ($^{\circ}$ C).

Experiment number	Protein concentration	Salt concentration	Rotation speed	Temperature	Optimized α_{max}	Optimized E	KL divergence
1	3	1.7	200	20	4.54E-07	0.01	13.04
2	2	1.7	200	20	6.67E-07	0.14	17.86
3	5	1.5	200	20	5.02E-07	0.41	8.95
4	2	1.5	198	20	4.24E-07	0.01	8.68
5	10	1.5	200	20	3.51E-08	0.38	12.80
6	2	1.5	199	20	2.29E-07	0.57	23.87
7	5	2	300	20	4.83E-08	1.05	10.86
8	5	2	100	30	3.66E-07	1.01	8.79
9	10	2	100	20	1.80E-07	0.02	7.44
10	5	2	300	30	2.49E-07	0.90	9.55
11	10	2	300	30	6.49E-08	0.78	9.00
12	10	1	100	20	4.54E-07	0.92	14.74
13	10	1	300	20	3.56E-08	0.94	11.846
14	10	1	300	30	6.49E-08	0.78	13.65
15	5	2	100	20	1.46E-07	0.01	9.72

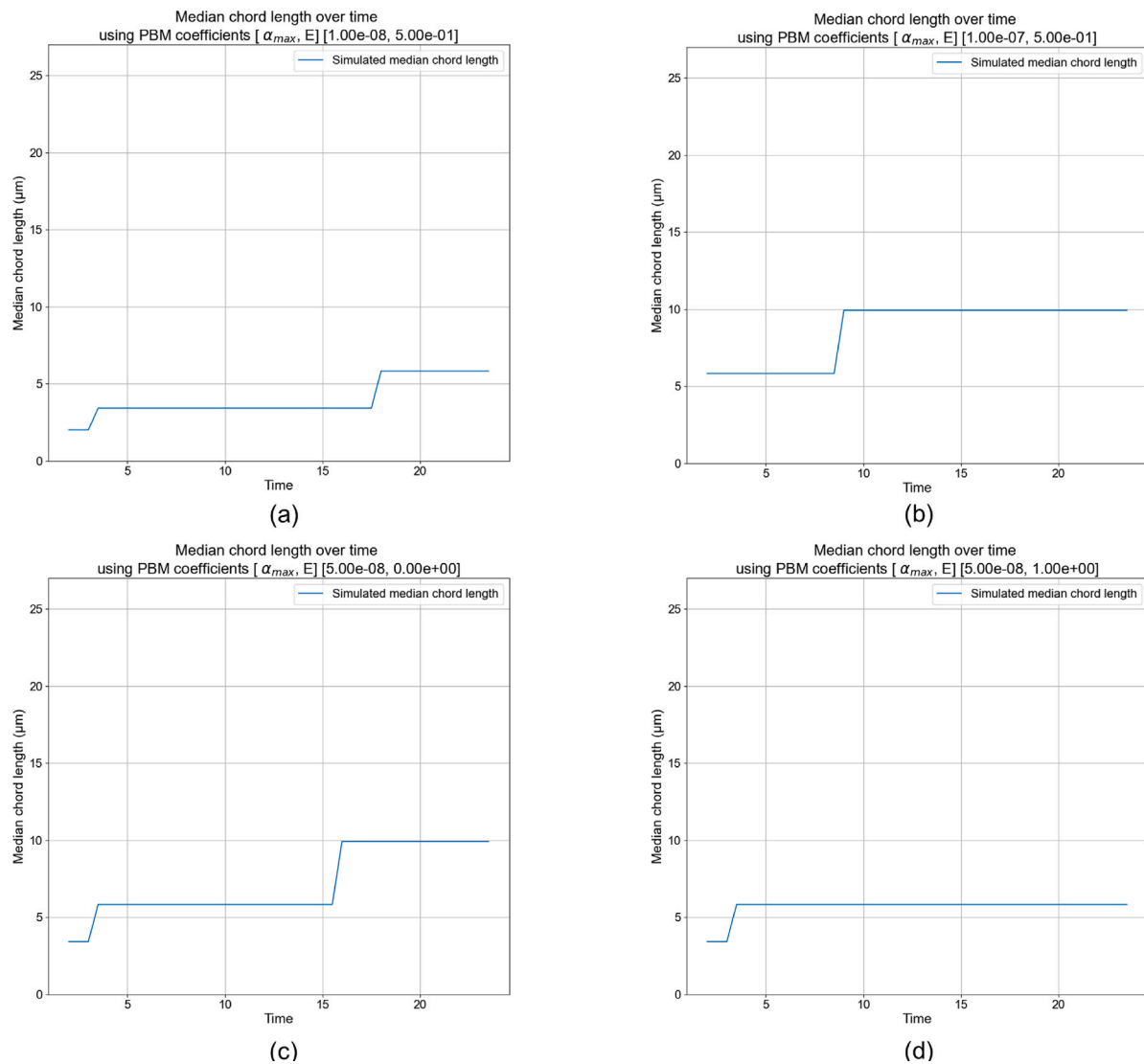


Fig. 8. Particle distribution median predictions to understand the effect of PBM parameters on the median; rotation speed for all of the models is considered 200 rpm. (a–b) Two cases in which the increase of efficiency is leading to the appearance of larger particles. (c–d) An increase in breakage coefficient will lead to smaller particles and also mathematical instability when the coefficient is near 3 or above.

from FBRM are discrete points and smooth curves representing the simulated distributions were obtained by interpolating the data onto a finer grid and applying a Gaussian filter to reduce noise, ensuring clear and accurate trends for visualization.

The results presented in Fig. 10 are the PSD using the regressed PBM parameters. Note that in Fig. 10 the y-axis is chosen to be normalized particle counts which reflect this work's aim in predicting the precipitation trend rather than particle counts directly. The range of the x-axis has been chosen because of the physical limitation of the Focused Beam Reflectance Measurement (FBRM) device, which cannot detect particles smaller than one micrometer. While the device cannot measure sub-micron particles, it is sufficient for precipitation processes as we expect them to be within this detectable range. The experiments that showed relatively good predictions from the model and had unique optimized solutions compared to other experiments were chosen. Even though the distributions are plotted from 1 h after starting the experiment, in some experiments, the FBRM is giving inaccurate measurements for very large particles, and the reason is the bubbles formed in the beginning of the experiment, which will disappear as time passes. It is

visible from the comparison of both distributions that the FBRM, as expected, is giving a wider distribution because it is counting some particles multiple times since it counts particles by a chord length, which in some cases might not be the diameter of the particle. The regions in which the PBM is giving almost perfect predictions are near both ends of the x-axis, while giving relatively okay predictions in the middle of the x-axis. Results in Fig. 11 are the median of the PSD from experimental and simulation data. It is a good indicator of how good the simulation is at representing the experimental data. The error bars on the measurements correspond to the FBRM maximum specified device error for median measurements, which is 4%. Experiment 9 shows the best match between the model and the experiment, and it can also be noted that it has the lowest objective function value out of these experiments. The model is over predicting for experiment 1 and at the end of experiment 4 but it is successfully capturing the trend of the experiment. The model performed relatively well in most of the experiments. The rest of the modeling results are given in the supplementary material (Section S4).

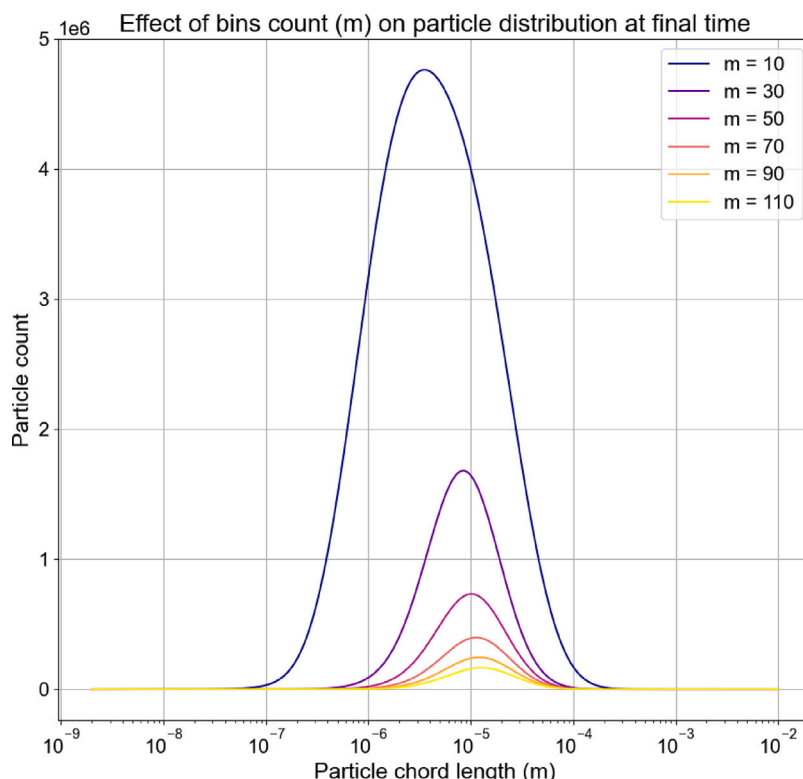


Fig. 9. Particle distribution after 48 h for experiment 1 with different bin counts.

4.4.4. Total volume of particles and the total number of particles for PBM

There were in total 15 experiments to be used in the solver/optimization algorithm to find the model parameters. Not all of the models performed similarly in all experiments.

As seen in Figs. 12 and 13 the total number and volume of particles are important for understanding the system and how different phenomena change the initial distribution.

The total number of particles should go down as expected from having agglomeration, and the total volume of particles should be constant. The total volume of particles is almost constant for all of the experiments, but there is a slight reduction of it, which might be because of the approximations of the breakage distribution function that was modified for a geometric discretization.

4.5. Step 6: Surrogate development and application

The optimized PBM parameters from step 5 and the experimental operating conditions are used to train an ANN model. The results of the surrogate model predictions and the predicted PSD are discussed in the following sections.

4.5.1. Predicting the population balance model parameters from process operating conditions

From Fig. 7 the effects of model parameters on the PSD has been explored in which an increase or decrease in those parameters will lead to the significance of either agglomeration or breakage which then will determine the formation of smaller or larger particles. A surrogate is used to model the relation between process operating conditions and the PBM. The results of the surrogate model can be summarized in Table 5. The model has been trained, and the outputs are averaged over 5 folds(each fold corresponds to a selection of some of the data as training sets and some as validation) and 10 000 epochs

each. The training and validation loss is provided in the Supplementary material(Fig. S2).

After the surrogate model had been trained to understand the relation between the process operating conditions such as protein and salt concentration, temperature and rotation speed (inputs) and the optimized parameters such as breakage rate coefficient and maximum collision efficiency (outputs), a nominal condition was chosen, and the predictions of the model parameters were done by keeping all but one process condition constant to see the effect of changing the process operating conditions on the optimized model parameters.

The predictions related to maximum collision efficiency $\alpha_{\max}$ are shown in Fig. 14 by varying temperature, salt concentration, rotation speed, and protein concentration. We see that the $\alpha_{\max}$ changes are not in multiple orders of magnitudes, and increasing temperature, rotation speed, and protein concentration seems to have an effect of lowering the maximum collision efficiency (Fig. 14(a,b,c)), in contrast to increasing salt concentration, which increases the collision efficiency. Increasing salt concentration increases the surface charges of lysozyme particles, creating a higher amount of physical forces aiding in more successful collisions. In contrast, increasing the rotation speed increases the shear forces between particles, in turn lowering the efficiency of collisions. The figures show signs of overfitting, which means more experimental data is needed for a more generalized model.

The effect of these operational parameters on the breakage rate coefficient, E , is shown in Fig. 15. The trends of result vary from the those of $\alpha_{\max}$. In Fig. 15(a,c), it can be seen that the temperature and rotation speed have a significant effect on the breakage rate coefficient. Fig. 15(b) shows that the breakage rate coefficient is not sensitive as much to salt concentration and the changes are very minimal. 15(c) shows that the breakage rate coefficient is increasing with increasing rotation speed, which can be explained as a result of increasing shear between the particles. Lastly, Fig. 15(d) shows that the protein concentration has little effect on the breakage rate coefficient. While the results

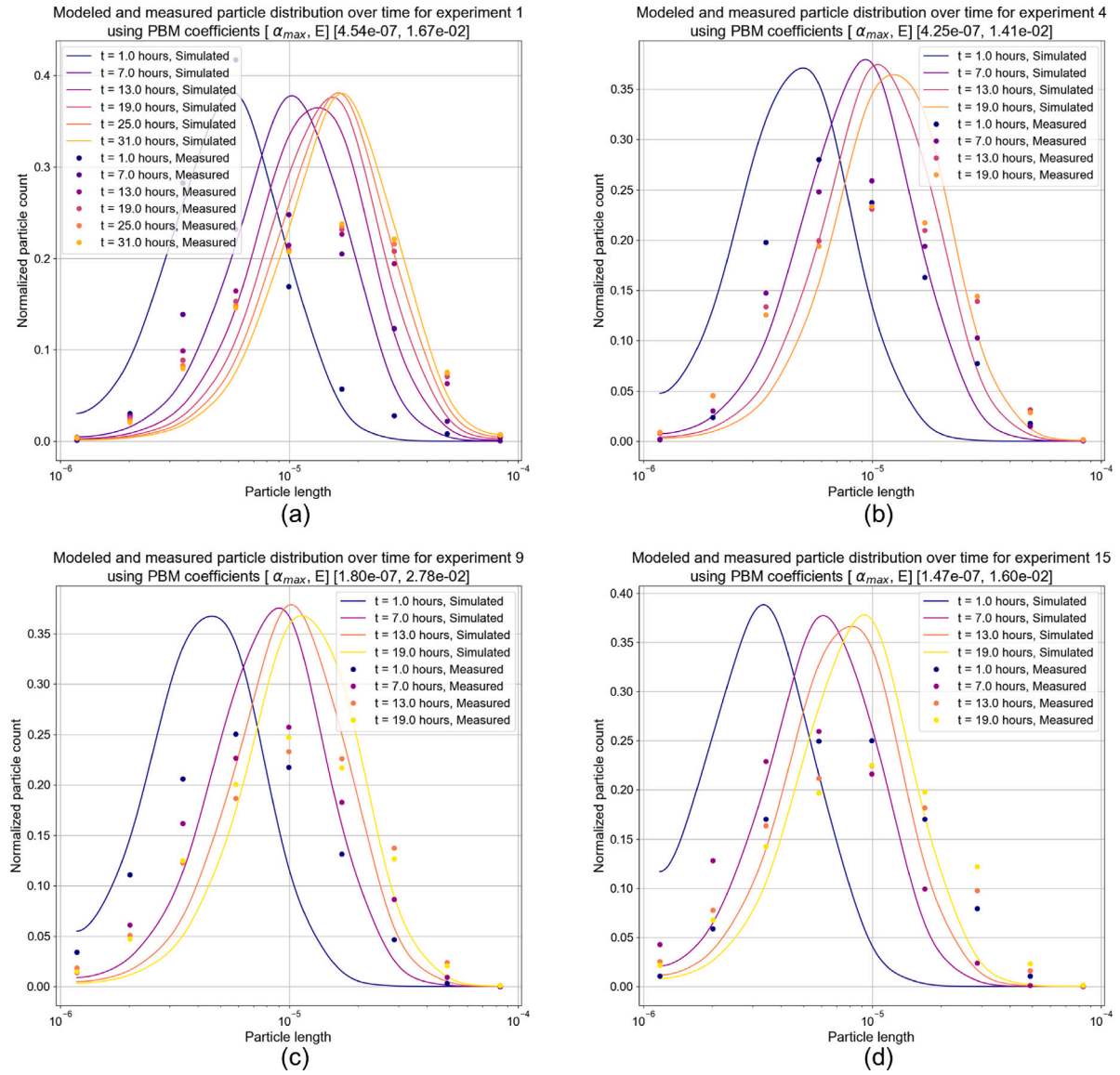


Fig. 10. Measured and simulated PSD for selected experiments using regressed PBM parameters from step 5. (a) PSD for experiment 1 with protein concentration of 3 g/L, salt concentration of 1.7 M, rotation speed of 200 rpm and temperature of 20 °C. KL divergence value of 13.04. (b) PSD for experiment 4 with protein concentration of 2 g/L, salt concentration of 1.5 M, rotation speed of 198 rpm and temperature of 20 °C. KL divergence value of 8.68. (c) PSD for experiment 9 with protein concentration of 10 g/L, salt concentration of 2 M, rotation speed of 100 rpm and temperature of 20 °C. KL divergence value of 7.44. (d) PSD for experiment 15 with protein concentration of 5 g/L, salt concentration of 2 M, rotation speed of 100 rpm, and temperature of 30 °C. KL divergence value of 9.72.

Table 5

An overview of the predictions of ANN for PBM parameters for each optimized experiment. The units for protein concentration (g/L), Salt Concentration (M), rotation speed (rpm), and temperature (°C). The surrogate model achieved an R^2 value of 0.88 for α_{max} predictions and 0.66 for E predictions.

Exp. number	Protein concentration	Salt concentration	Rotation Speed	Temperature	Optimized α_{max}	Optimized E	Predicted α_{max}	Predicted E
1	3	1.70	200	20	4.54e-07	0.0167	4.77e-07	0.09
2	2	1.70	200	20	6.67e-07	0.1429	6.35e-07	0.00
3	5	1.50	200	20	5.02e-07	0.4170	4.20e-07	0.23
4	2	1.50	198	20	4.24e-07	0.0141	3.32e-07	0.25
5	10	1.50	200	20	3.51e-08	0.3825	7.12e-08	0.61
6	2	1.50	199	20	2.29e-07	0.5718	3.30e-07	0.25
7	5	2.00	300	20	4.83e-08	1.0534	5.37e-08	0.72
8	5	2.00	100	30	3.66e-07	1.0191	3.22e-07	0.62
9	10	2.00	100	20	1.80e-07	0.0278	1.55e-07	0.14
10	5	2.00	300	30	2.49e-07	0.9063	2.17e-07	0.78
11	10	2.00	300	30	6.49e-08	0.7830	9.56e-08	0.75
12	10	1.00	100	20	4.54e-07	0.9243	4.00e-07	0.48
13	10	1.00	300	20	3.56e-08	0.9441	4.42e-08	0.78
14	10	1.00	300	30	6.49e-08	0.7830	6.35e-08	0.75
15	5	2.00	100	20	1.46e-07	0.0160	3.20e-07	0.06

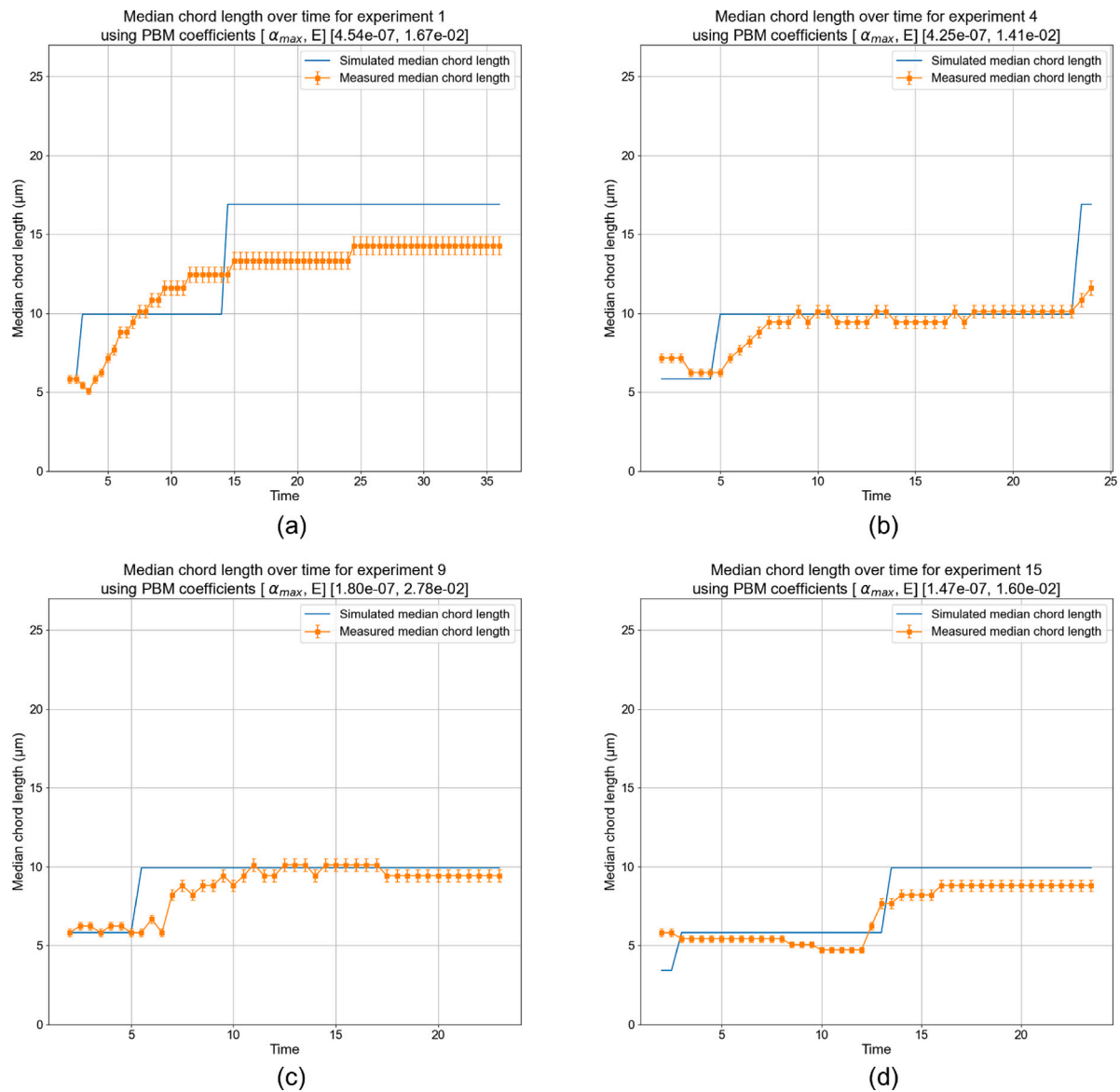


Fig. 11. Median of measured and simulated PSD for selected experiments. (a) Experiment 1 with protein concentration of 3 g/L, salt concentration of 1.7 M, rotation speed of 200 rpm and temperature of 20 °C. KL divergence value of 13.04. (b) Experiment 4 with protein concentration of 2 g/L, salt concentration of 1.5 M, rotation speed of 198 rpm and temperature of 20 °C. KL divergence value of 8.68. (c) Experiment 9 with protein concentration of 10 g/L, salt concentration of 2 M, rotation speed of 100 rpm and temperature of 20 °C. KL divergence value of 7.44. (d) Experiment 15 with protein concentration of 5 g/L, salt concentration of 2 M, rotation speed of 100 rpm, and temperature of 30 °C. KL divergence value of 9.72.

obtained in this segment might suffer from underfitting, the variance in both parameter values shows little variance with salt concentration and protein concentration but a larger variation with temperature and rotation speed. Having these predictions of PBM parameters based on process operating conditions gives us the ability to gain a prediction of the output when we are designing the experiments, and the predictions work as a good initial guess for the optimization loops.

4.5.2. Particle size distribution from the PBM parameters predicted by the surrogate model

The data from Table 5 is used to solve the PBM with the predicted PBM parameters, a contrast to solving the PBM with the optimized PBM parameters. All of the results of using predicted PBM parameters are given in the supplementary material (section S5). Comparing Figs. 10 to 16, we observe similarity between the PSD predicted with the regressed PBM parameters in Step 5, and PSD predicted with the predicted parameters using the surrogate model in Step 6. The changes are better visualized in comparing Figs. 8 to 17, the results are similar

although the differences can be seen, but for example experiment 4, the surrogate model does not have the median jump at the end of the experiment. These results show that the surrogate model successfully predicts the trend of the precipitation and can be used to give predictions close to the measured experimental data and to be used to design new experiments for future work.

5. Conclusion

In this work, we have explored dynamical modeling and ML to elucidate the mechanisms of protein precipitation from process operating conditions. The PBM model allows us to create a dependence between its critical parameters to the measured PSD. However, due to the computational and experimental effort to obtain these parameters, we have presented a surrogate modeling strategy to predict the PBM parameters from the process operating conditions data. We have also shown the use of measured distributions from the FBRM probe for the amount of particles in a length scale to get regressed PBM parameters. Combining a

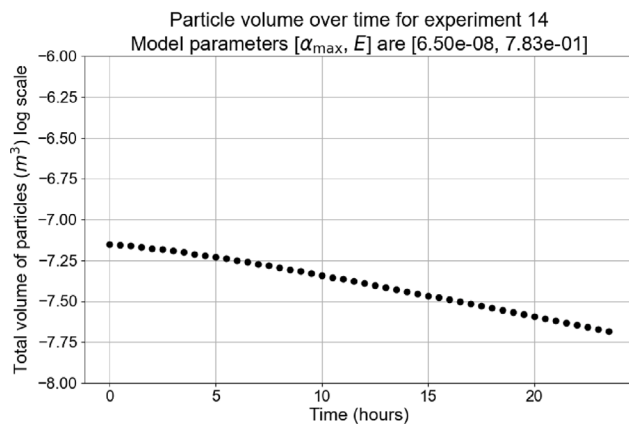


Fig. 12. Total volume of particles over time which is an indirect indicator of conservation of mass, the time unit is hours. Note the scale of the y-axis is logarithmic so that the deviation of the particle volume is negligible on this scale. Thus a constant value.

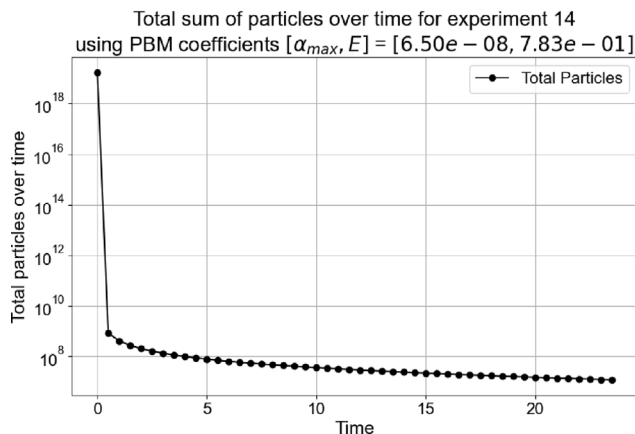


Fig. 13. Total particles over time for experiment 14, since agglomeration and particle growth are the dominant phenomena, the number of particles should decrease. The time unit is hours.

mechanistic model for PBM given the critical process model parameters of maximum collision frequency ($\alpha_{\max}$) and breakage coefficient (E), with a surrogate model to correlate the operational data temperature, rotation speed, protein concentration, salt concentration, showcases a foundation for future work to realize a hybrid, mechanistic and data-driven model. In this study, the modeling approach has been presented to address the recovery of proteins like lysosomes. PBMs are utilized to predict PSDs, and process operating conditions are correlated with PBM parameters, in this work. The combination of PBM with a surrogate model (ANN) helps find the parameters of the mechanistic dynamical model. The surrogate model also aids in quantifying mechanistically-plausible dependencies of the process operating conditions on PBM parameters of interest.

While in this work KL divergence was used to regress the PBM parameters to PSD, other approaches could involve using other objective functions like energy distance and maximum mean discrepancy, to relate FBRM probe measurements to the number of particles rather than just the distribution. Future work could also consider additional process operating conditions, tuning additional PBM parameters for better accuracy. A global sensitivity analysis could help identify the most influential parameters while addressing measurement errors, and prediction uncertainties would improve the model's reliability.

While the approach is generalizable for any protein with similar properties to lysozyme, we observe that the main changes to the model will relate to fine-tuning the correlations and the ML model. We note that the amount of data required for a generalizable model was not realized for the ML model, as some spurious correlations were extrapolated by the model. To alleviate this, we showed how creating different folds of cross-validated data can be used to train a surrogate model for the correlation between the process operating conditions and the PBM parameters. Further, the optimized values of parameters for collision efficiency and breakage rate coefficient suffer from suboptimal training. Despite this limitation, certain correlations obtained through the surrogate approach adhere to a mechanistically plausible interpretation, lending credibility to the model and the modeling framework. These limitations with the modeling can be subverted with accelerated computing and global optimization strategies that require more information on the PBM parameters.

Nomenclature

Abbreviations

DSP	Downstream processing
DE	Differential evolution
FBRM	Focused Beam Reflectance Measurements
FFANN	Feed forward artificial neural network
ML	Machine learning
PBM	Population balance model
PVM	Process Video and Measurement
PSD	Particle size distribution

Latin Letters

B	Birth terms
D	Death terms
D_f	Fractal dimension
d	Vessel inner diameter
E	Breakage rate constant
G	Average velocity gradient
h_1	Height of the stirrer
L	Particle length
L_p	Primary particle length
N	Particle density function (discretized)
R	Geometric step size
S	Breakage rate
V_L	Vessel filled volume
x	Bin edges after discretization
u	Agglomerate or particle volume
t	Time
Ne'	Modified power number
Re	Reynolds number
$\omega(u)$	Angular velocity of the stirrer
ν	Kinetic viscosity

Greek Letters

α	Collision efficiency
$\alpha_{\max}$	Maximum collision efficiency
β	Collision frequency
Δ	Half of the bin lengths
δ	Kronecker delta
ϵ	Homogeneous turbulent energy dissipation rate of the mixed tank
η	Dynamic viscosity
$\eta_{i,j,k}$	Proportionality coefficient
γ	Breakage distribution function
ρ_c	Density of solution
σ	Length scale
ν	Agglomerate or particle volume
ν	Kinetic viscosity

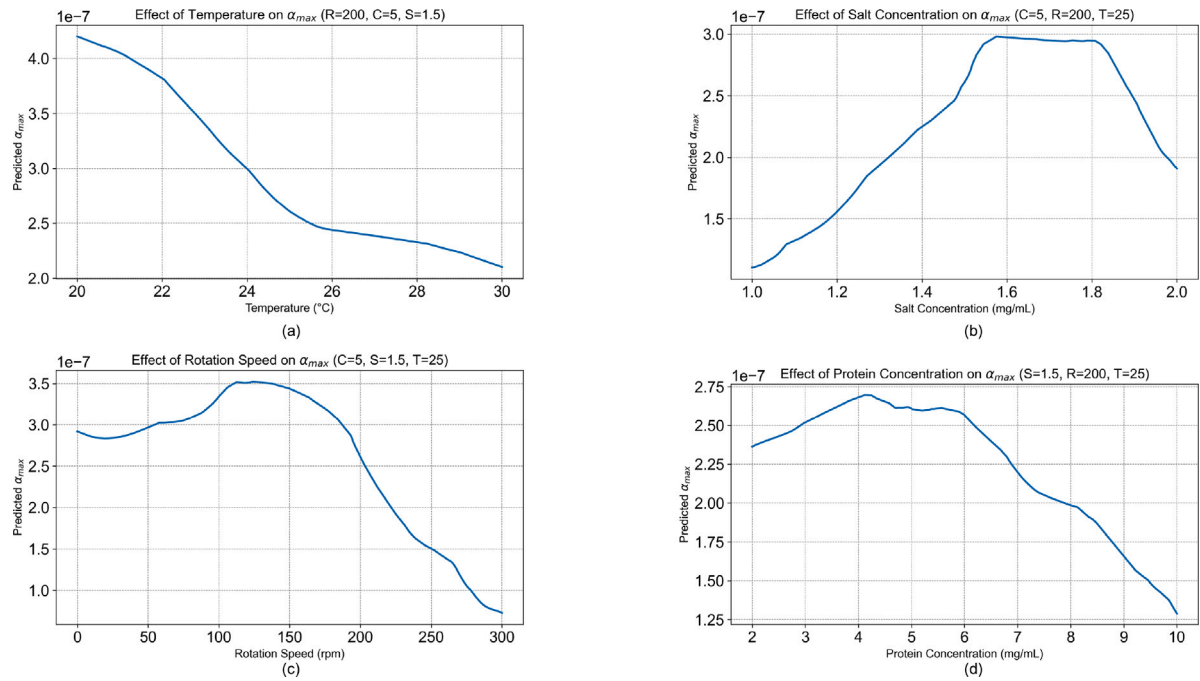


Fig. 14. Neural network predictions of model parameters based on process operating conditions for α_{max} .

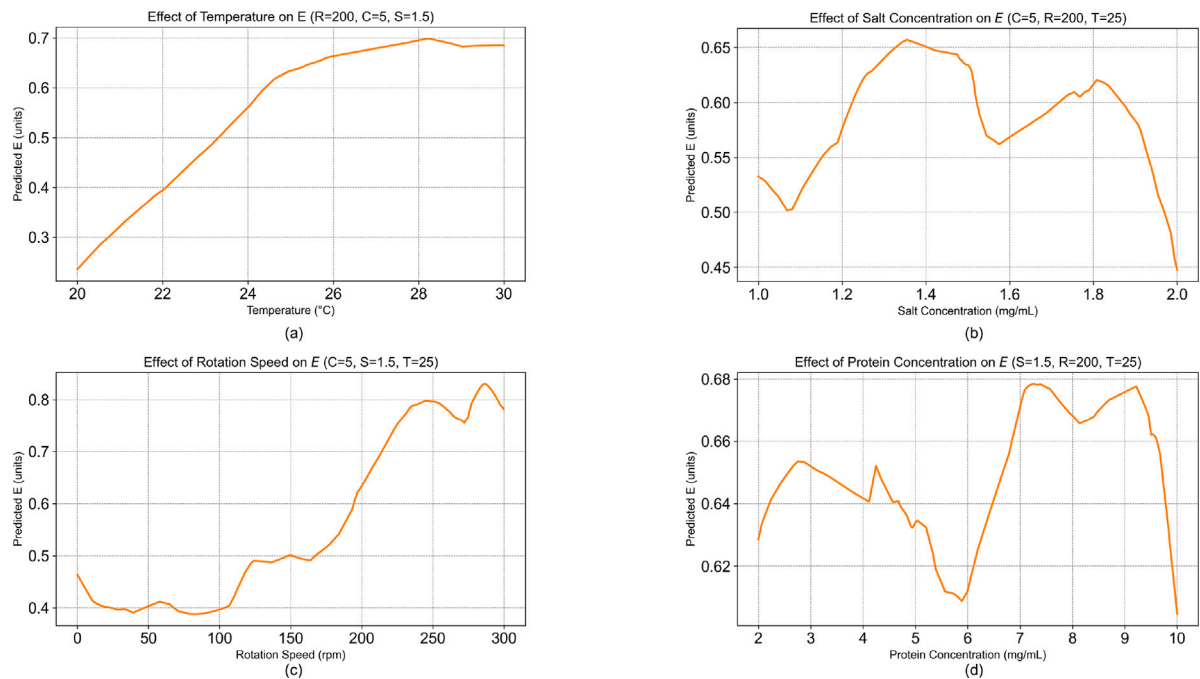


Fig. 15. Neural network predictions of model parameters based on process operating conditions for E .

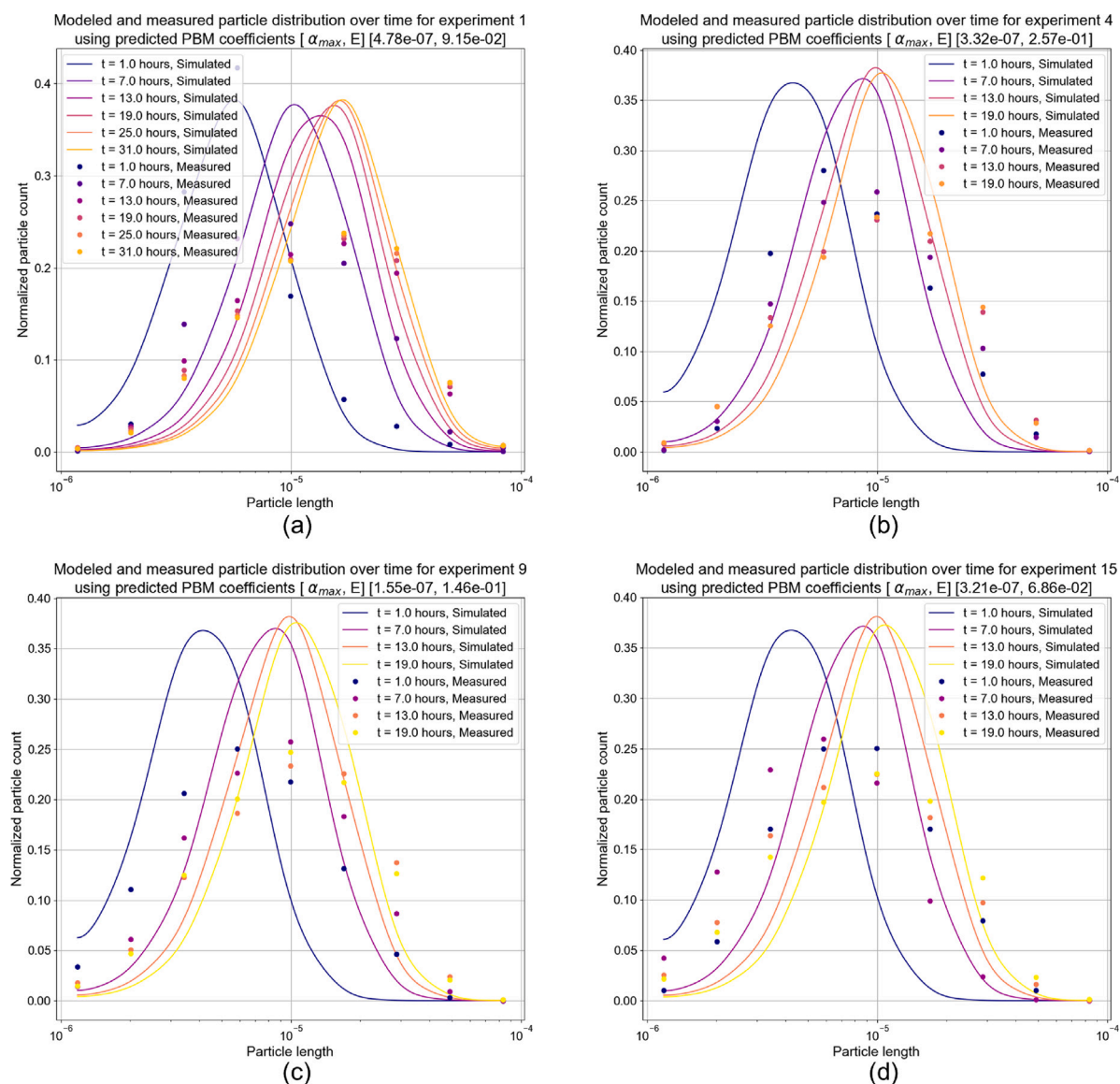


Fig. 16. Measured and simulated PSD for selected experiments with the predicted parameters from the surrogate model and using them in the PBM. The process operating conditions for each experiment were provided to the surrogate model, to estimate the PBM parameters which are used to simulate the PBM (a) PSD for experiment 1 with protein concentration of 3 g/L, salt concentration of 1.7 M, rotation speed of 200 rpm and temperature of 20 °C. (b) PSD for experiment 4 with protein concentration of 2 g/L, salt concentration of 1.5 M, rotation speed of 198 rpm and temperature of 20 °C. (c) PSD for experiment 9 with protein concentration of 10 g/L, salt concentration of 2 M, rotation speed of 100 rpm and temperature of 20 °C. (d) PSD for experiment 15 with protein concentration of 5 g/L, salt concentration of 2 M, rotation speed of 100 rpm, and temperature of 30 °C.

CRediT authorship contribution statement

Amirkiarash Ehtesham: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Abhishek Sivaram:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Sara Danielle Siegel:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **John van Zanten:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Sayed Soheil Mansouri:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compchemeng.2025.109193>.

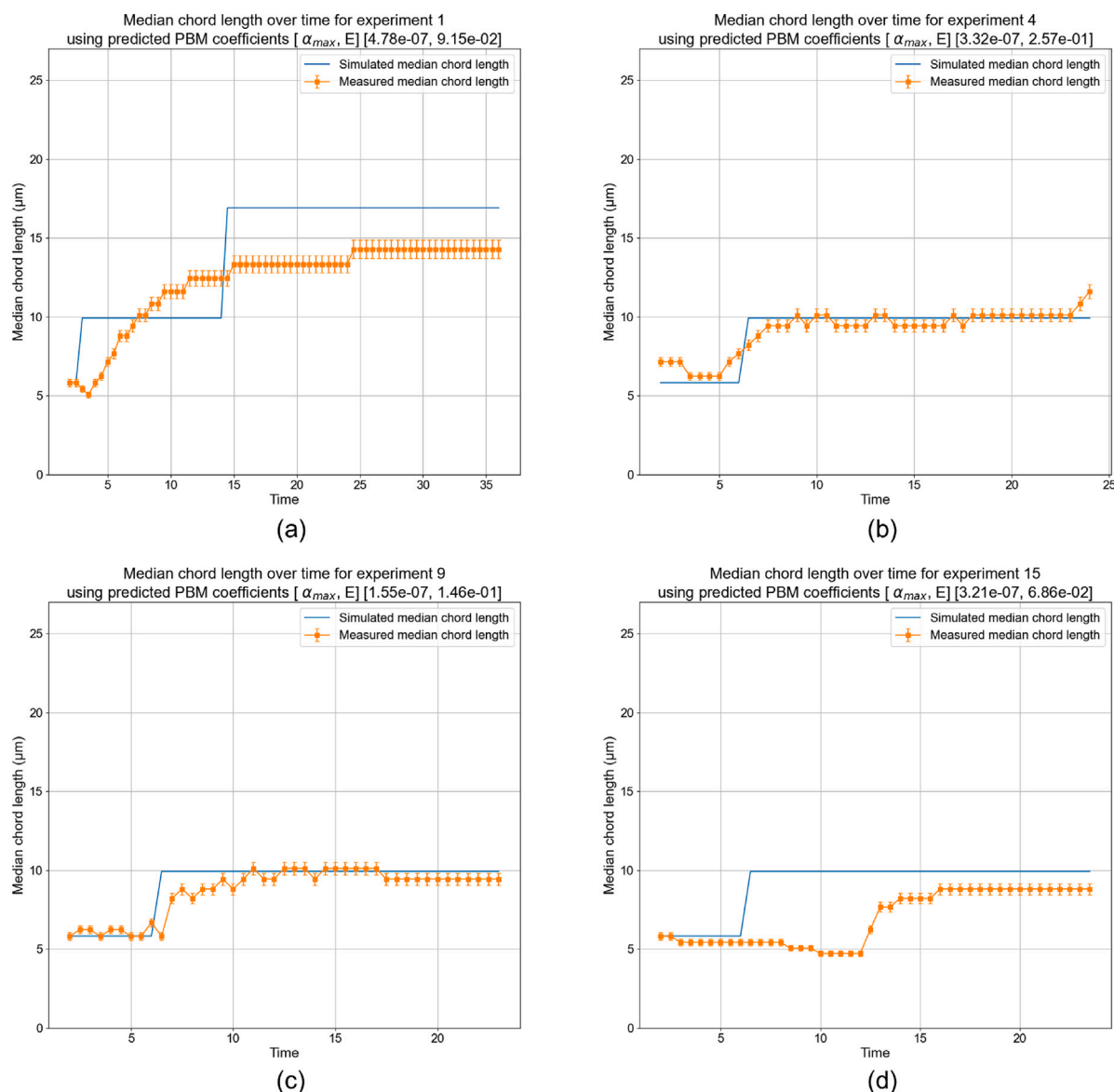


Fig. 17. Median of measured and simulated PSD for selected experiments with the predicted parameters from the surrogate model. (a) Experiment 1 with protein concentration of 3 g/L, salt concentration of 1.7 M, rotation speed of 200 rpm and temperature of 20 °C. (b) Experiment 4 with protein concentration of 2 g/L, salt concentration of 1.5 M, rotation speed of 198 rpm and temperature of 20 °C. (c) Experiment 9 with protein concentration of 10 g/L, salt concentration of 2 M, rotation speed of 100 rpm and temperature of 20 °C. (d) Experiment 15 with protein concentration of 5 g/L, salt concentration of 2 M, rotation speed of 100 rpm, and temperature of 30 °C.

Data availability

Data will be made available on request.

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